Supporting Information for: What Cycle Closure Can and Cannot See in a Relative Binding Free Energy Network

Nikita L. Polomoshnov
Faculty of Bioengineering and Bioinformatics, Lomonosov Moscow State University, Moscow, Russia
Institute of Biomedical Chemistry (IBMC), Moscow, Russia
nikitapol@fbb.msu.ru

August 29, 2026

Supporting Information

Reproducibility. The analysis code released with this work regenerates every figure in this article, each from a single command, and it emits the per-system tables reported here from the same run that draws the corresponding figure, so the two cannot drift apart. This covers the quality-control results (Figures 7 and 3 of the main text and Figures S12, S13 and S14 here), the observability map of Figure S9, the fixed-cutoff comparison of Figure S11, the σ-profile sweeps of Figures S15 and S16, the autocorrelation panel of Figure S1, the curated-label grounding of Figure S6 and Table S3, the label-noise floor of Figure S7, and the cross-release split of Figure S8. The constants appearing in the theorems and the invariants of the cycle-closure test are additionally asserted as unit tests that run in the same environment. Every input is public: the alchemtest BACE1 and benzene sets, the OpenFF protein–ligand benchmark, ChEMBL [22], and the OpenFE IndustryBenchmarks2024 replicate set [3]. Recalibration baselines use temperature scaling [9] and isotonic regression [21].

BACE1 per-window overlap values (Figure 4). Figure 4 reports that naive $1/I$ overstates the bootstrap se by 2–12× on the 19 real BACE1 adjacent-λ BAR windows (mean 6.07, $[2.78, 12.01]$), the worst overstatement occurring at high overlap, consistent with the controlled panel (invariant #1, main text). These 19 rows are the windows of one λ schedule, taken from the complex leg of the alchemtest AMBER `load_bace_example` relative-binding calculation across its decharge, recharge and van der Waals legs; each is a two-state BAR estimate between neighbouring λ values, so the set tests the closed-form variance against a decorrelated bootstrap on real molecular-dynamics works, and it does not test 19 separate binding free energies or reach beyond the high-overlap band this schedule occupies. Table S1 lists the (λ_i, λ_j) window labels, per-window overlap $4\langle p(1-p)\rangle$, and both ratios for all 19, emitted verbatim by the released analysis code, λ labels included, so the λ pairing is re-derivable from it (seed 20260629; both sandwich inputs and the bootstrap truth subsampled to the statistical inefficiency g before B is formed, Section 5; numerically identical to the binding rows of the corresponding panel). Every window sits at overlap ≥ 0.84 (mean 0.93); the worst naive/bootstrap ratio (12.01×) occurs at the single highest-overlap window ($4\langle p(1-p)\rangle = 0.992$), and the best (least overstated, 2.78×) occurs at the lowest-overlap window in the set (0.846), the same high-overlap-is-worst regime as the controlled sweep of Figure S2 rather than a property of the particular windows sampled. The trend across the full 19-window set is not confined to the

Table S1: **Per-window overlap on the 19 real BACE1 adjacent-λ BAR windows.** λ_i, λ_j are the BAR window endpoints; $4\langle p(1-p)\rangle$ is the normalized BAR overlap; sandwich/boot and naive/boot are the reported-se/bootstrap-se ratios for the sandwich B/I^2 and the naive $1/I$, respectively, against a decorrelated-bootstrap reference standard error. Rows are sorted by overlap, ascending; repeated (λ_i, λ_j) pairs are distinct windows from BACE1's decharge, recharge, and van der Waals alchemical legs, which share a common λ schedule.

λ_i	λ_j	$4\langle p(1-p)\rangle$	sandwich/boot	naive/boot
0.437	0.563	0.846	0.992	2.780
0.316	0.437	0.858	1.001	3.020
0.206	0.316	0.876	0.967	4.377
0.750	1.000	0.884	0.956	3.315
0.563	0.684	0.909	1.041	3.451
0.500	0.750	0.917	0.971	3.618
0.684	0.794	0.921	0.997	3.581
0.250	0.500	0.927	1.017	3.860
0.000	0.250	0.928	0.967	3.678
0.500	0.750	0.943	1.018	9.823
0.794	0.885	0.943	0.981	4.165
0.000	0.250	0.948	0.986	9.360
0.115	0.206	0.949	1.005	6.414
0.250	0.500	0.967	1.004	9.116
0.750	1.000	0.968	1.052	9.901
0.885	0.952	0.969	0.995	5.670
0.952	1.000	0.983	1.064	8.193
0.048	0.115	0.984	1.066	8.960
0.000	0.048	0.992	1.007	12.007

extremes: the Spearman rank correlation between overlap and the naive/bootstrap ratio is $\rho = 0.84$ ($n = 19$, computed from Table S1), so the relationship holds within the narrow $[0.846, 0.992]$ overlap band, not just at its endpoints.

Autocorrelation robustness: raw n vs $n_{\text{eff}} = n/g$ (Figure S1). The sandwich $B = n_f \operatorname{Var}_f[p] + n_r \operatorname{Var}_r[p]$ (Theorem 3) assumes independent samples, whereas real molecular-dynamics works are autocorrelated. On the 19 real BACE1 adjacent-λ BAR windows of Figure 4 (no new molecular dynamics), the sandwich se is recomputed at the raw sample count and again after decorrelating each leg to its statistical inefficiency g [18] (geometric-mean $\bar{g} = 1.48$, range $[1.15, 2.16]$; median count n=500 $\rightarrow$ 349). Against the correlation-aware truth (the decorrelated bootstrap standard deviation, the same reference as Figure 4), the n_{eff} sandwich is calibrated, at a reported/truth ratio of 1.00 $[0.93, 1.07]$, reproducing Figure 4. The raw-n sandwich is overconfident by 0.83× $[0.67, 0.97]$, matching the $1/\sqrt{\bar{g}} = 0.82$ autocorrelation prediction. The sandwich also tracks its own matched (raw) bootstrap (1.00, $[0.92, 1.05]$): the closed form is internally consistent, and the only quantity that must be right is the effective sample count, which the paper decorrelates to g throughout (Section 5, main text). The calibration verdict is therefore robust to autocorrelation once n_{eff} is used, as it is in every real-data panel.

The learned-variance comparison (Figure S2). Figure S2 is the controlled overlap sweep behind the comparison the main text states: the sandwich B/I^2 coincides with `pymbar`-MBAR and with Monte-Carlo truth across the range, while a fair learned mean–variance head, given the same

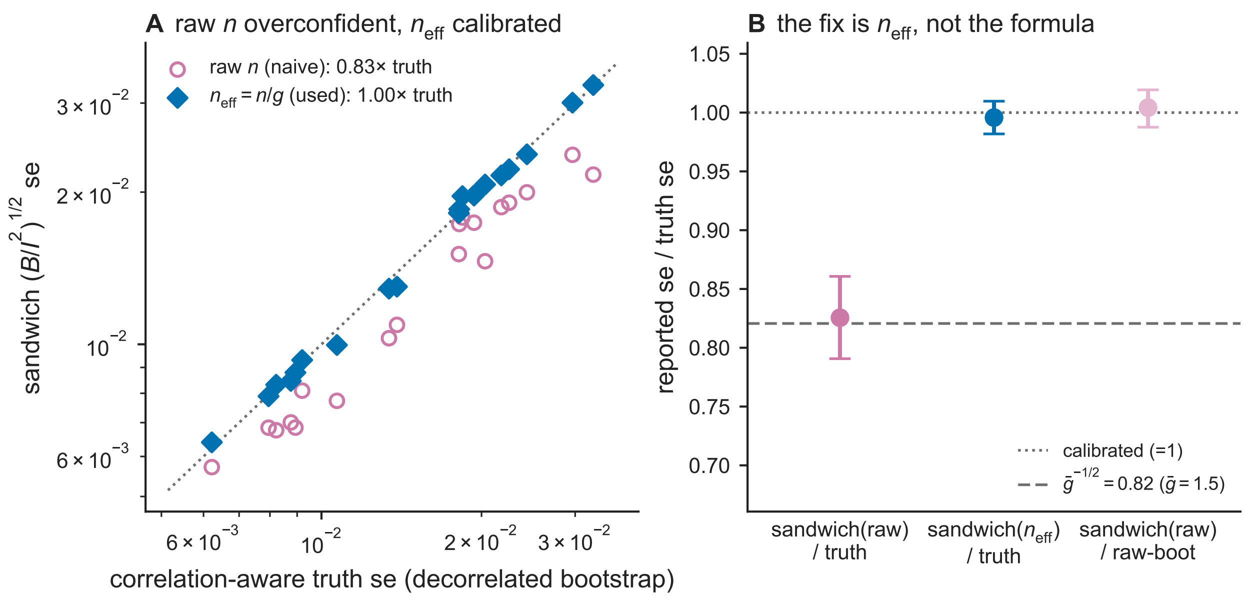

Figure S1: **Sandwich standard error at the raw sample count versus at the effective sample count $n_{\text{eff}} = n/g$, on the real BACE1 λ windows.** (A) Sandwich standard error at n_{eff} and at raw n, plotted against the correlation-aware (decorrelated bootstrap) reference standard error, with the $y{=}x$ line. (B) Aggregate ratio of sandwich standard error to the reference, at n_{eff}, at raw n, and of the sandwich to its own raw-n-matched bootstrap, with bootstrap confidence intervals; the $1/\sqrt{\bar{g}}$ autocorrelation prediction is marked.

overlap information the closed form uses, is overconfident at a realistic label budget and remains so at a twenty-fold larger one. Two restrictions travel with the figure. The head is trained and measured here, but its use as a comparison bar downstream is a counterfactual: no deployed or published pipeline is known to feed a learned per-edge σ into a network consistency check, and the arms built from its measured miscalibration in the quality-control and stopping experiments are comparison models rather than estimators run in those experiments. And the overconfidence is a property of the per-edge parameterisation rather than of learning from labels, since on the identical labels pooling single-label edges by overlap recovers the sandwich to $\sim$6% at 4000 edges (below).

Corrected learned-variance foils and the data-efficiency floor (Figure S2). Neither correction restores the head's calibration at a realistic budget. A head trained on the β-weighted negative log-likelihood (β-NLL) [17] (β=0.5, the standard objective correction) does not recover calibration (se $\approx$0.11$\times$, slightly worse than plain NLL's 0.15$\times$), and at this training seed a 5-member deep ensemble [13] reaches only the large-budget floor (se $\approx$0.20$\times$, the same $\approx$5$\times$ overconfidence a 20$\times$-larger-budget plain head reaches). The residual $\sim$5$\times$ overconfidence of the plain, oracle and β-NLL heads is therefore a property of the per-edge parameterisation rather than of the label budget or of the Gaussian-NLL objective. The conditional variance se(overlap) is identifiable from single labels, and pooling single-label edges by overlap recovers the sandwich to $\sim$6% at 4000 edges ($\sim$4% at 40000) on the same labels the heads were given. This bears on the scope of the quality-control comparison. Placing a 6% miscalibration on the frozen scale grid of Figure S16 gives seven flagged systems of 48 against the published six, and a 6% over-estimate gives six; the test is unchanged. The collapse to 42 of 48 requires the 5–11$\times$ under-estimate the trained per-edge heads produce, so the contrast is scoped to heads of that kind and is not evidence that a same-budget learned estimator

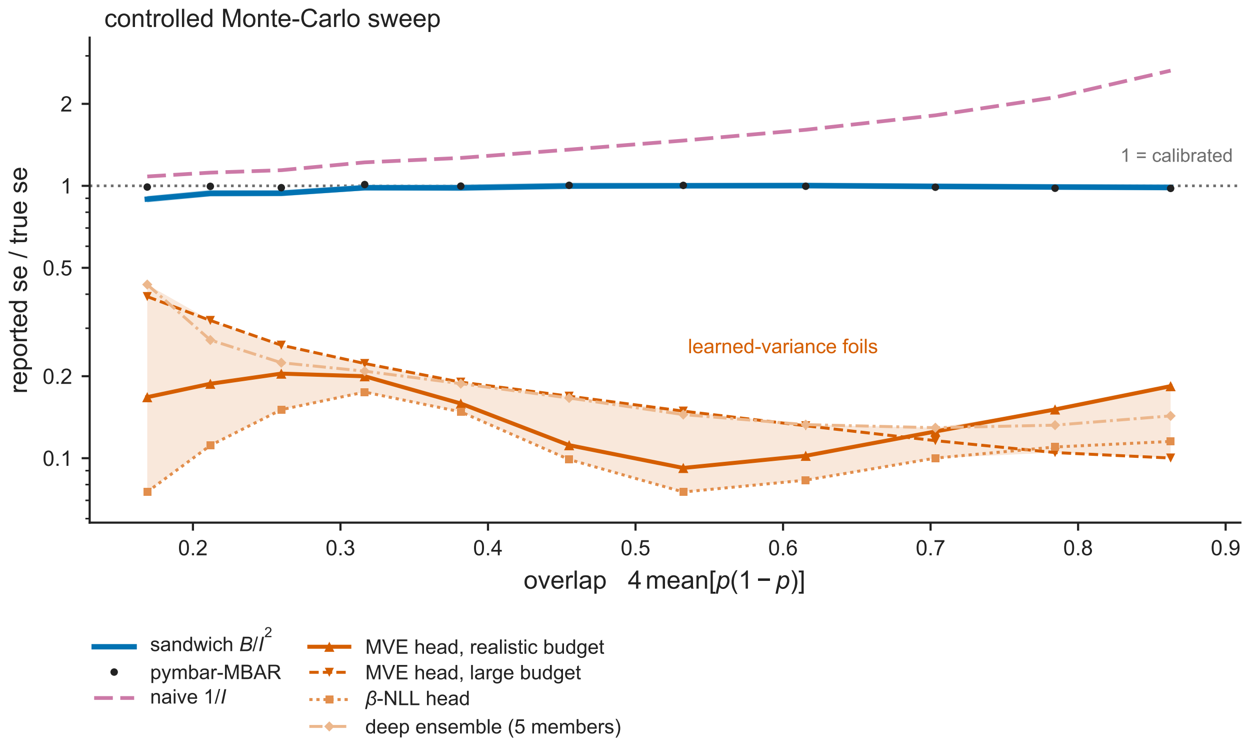

Figure S2: **Reported standard error against the true standard error over a controlled overlap sweep.** Reported standard error divided by the true standard error, versus work-distribution overlap, for the sandwich B/I^2, `pymbar`-MBAR, the naive information-equality plug-in $1/I$, a learned mean–variance (MVE) head at a realistic and at a large training budget, a β-NLL head, and a 5-member deep ensemble, in a controlled Monte-Carlo sweep.

must break the test.

Seed stability of the learned-variance foils (Figure S2). These ratios are stable across independent training seeds for the plain and β-NLL heads: over 5 seeds the plain head spans 0.1229–0.2103$\times$ (mean 0.1537$\times$) and the β-NLL head 0.1024–0.1256$\times$ (mean 0.1147$\times$), so their overconfidence is a property of the estimator at this budget rather than of a single initialization. The 5-member ensemble is not stable: across the same 5 seeds, each with a disjoint 5-network member block, it spans 0.1984–1.2200$\times$ (mean 0.4198$\times$, sd 0.4481$\times$), and one seed overestimates the true se by 22%, crossing perfect calibration. Unlike the plain, oracle (0.1941–0.2571$\times$, mean 0.2101$\times$), and β-NLL heads, whose overconfidence is robust across independent retrains, the ensemble is unreliable in both directions at this budget.

Synthetic decomposition (illustration). In a controlled one-dimensional problem whose generative structure is known, the decomposed interval is up to 2.2$\times$ sharper than a split-conformal interval (2.0$\times$ sharper than conformalized quantile regression [15]) at equal coverage, with better conditional calibration. The problem illustrates the mechanism; the evidence on real data is Figure S18.

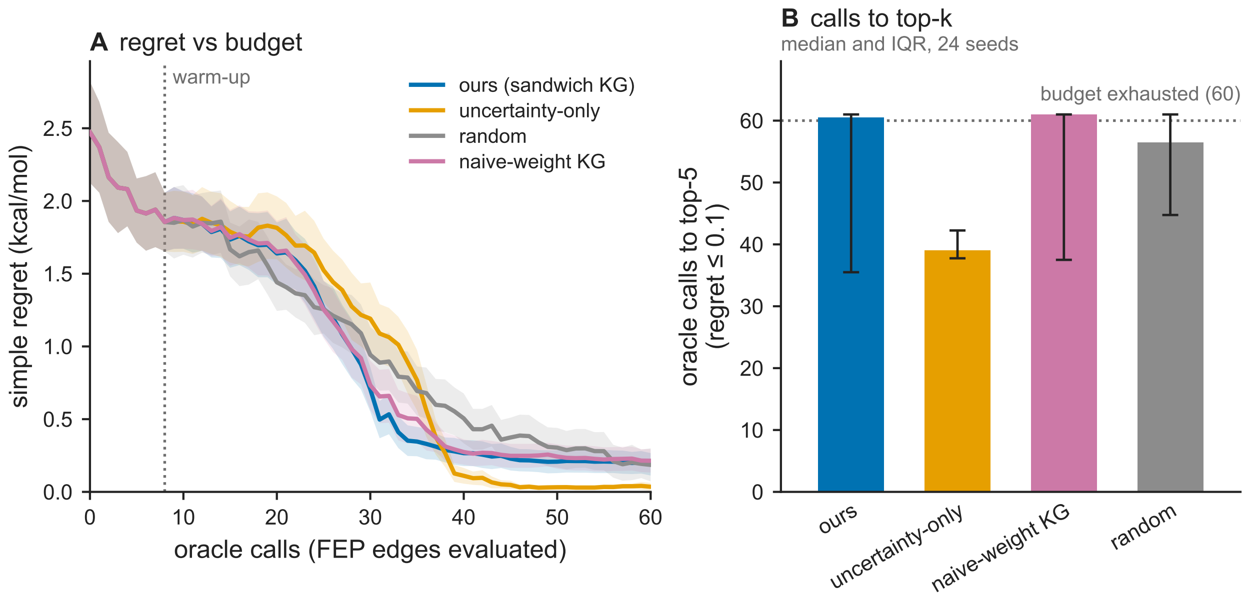

Figure S3: **Simulated active-learning acquisition policies.** (A) Simple regret versus oracle calls (simulated FEP edges evaluated), for a sandwich-weighted knowledge-gradient policy [7], implemented as in Balandat et al. [2] and Gardner et al. [8], a naive-weighted knowledge-gradient policy, an uncertainty-only policy, and random selection, averaged over 24 seeds with 95% confidence bands. (B) Median oracle calls to reach the top-k candidates (regret at or below tolerance), with interquartile-range error bars, for the same four policies.

Edge weighting (Figure S3). The acquisition policy outperforms random selection by a factor of 2.2. Changing the edge weighting from naive to sandwich changes calls-to-top-k by a factor of 0.98 $[0.93, 1.03]$ (24-seed bootstrap confidence interval), which is the Gauss–Markov ceiling on the choice of weights.

The Fisher–resistance correspondence. The theorem below establishes what the per-edge bar buys the decision layer: with the inverse sandwich variances as conductances, the Gaussian posterior precision of the node potentials is a weighted Laplacian, contrast variance is effective resistance, acquisition updates are $O(1)$, and the knowledge gradient is exactly zero on gauge directions. It is stated here because what the article measures downstream of it is that the weight *value* is second-order for ranking: in a connected graph the generalized-least-squares contrast mean is unbiased for any positive conductances, so by Gauss–Markov the choice of weights cannot improve a ranking (Table S11 below). The theorem's content is accordingly the gauge-invariance, which is structural and holds for any positive conductances, and which the simulation of Figure S4 exercises.

Theorem S1 (Fisher–resistance correspondence)**.** *For relative measurements $y_e = (\varphi_j - \varphi_i) + \varepsilon_e$, $\varepsilon_e \sim \mathcal{N}(0, V_e^{\text{true}})$ with $V_e^{\text{true}} = B_e/I_e^2$, the Gaussian posterior precision of the node potentials is the weighted Laplacian $L = \sum_e w_e(e_i - e_j)(e_i - e_j)^\top$ with conductances $w_e = 1/V_e^{\text{true}} = I_e^2/B_e$. Contrast variance equals effective resistance [12], a single edge recovers V_e^{true}, Sherman–Morrison gives $O(1)$ acquisition updates, and $\ker L \supseteq \operatorname{span}\{\mathbf{1}\}$, so the knowledge gradient (KG) [7] is exactly zero on gauge/redundant directions: the acquisition is automatically gauge-invariant.*

The edge weights are the inverse sandwich variances, the readout of Section 5 entering the decision layer; the acquisition model identifies the reported variance with the true one ($c_e = 1$), an

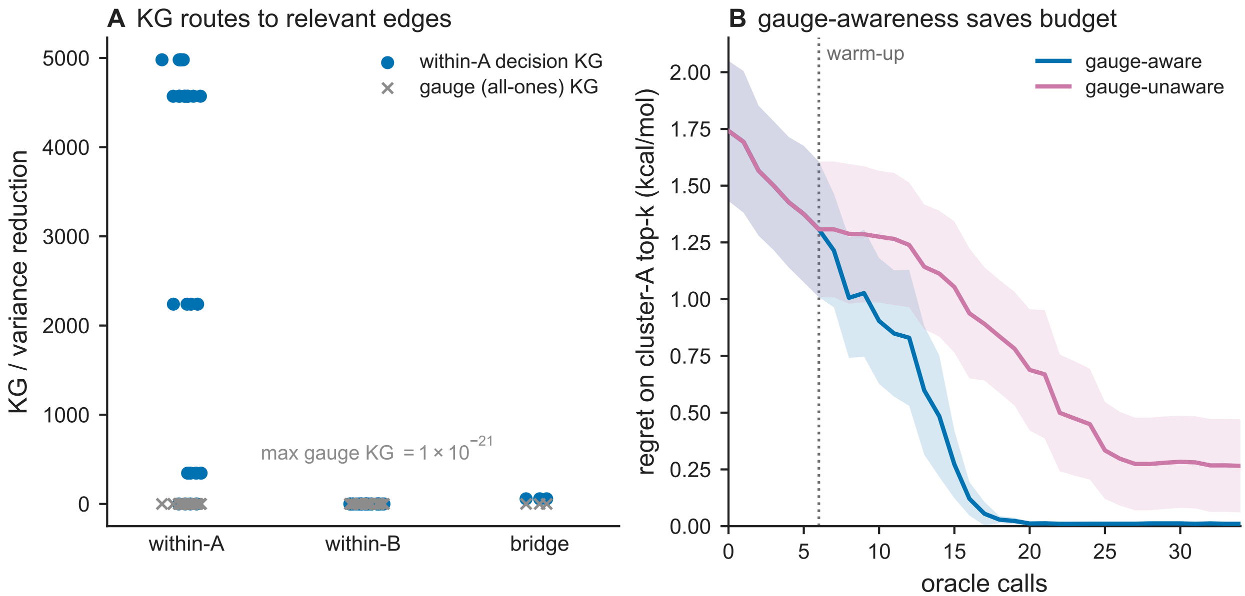

Figure S4: **Knowledge-gradient score on gauge and decision-relevant contrasts (simulation).** (A) Knowledge gradient for the within-cluster decision contrast and for the gauge (all-ones) contrast, by edge type (within-cluster-A, within-cluster-B, bridge). (B) Regret on the cluster-A top-k candidates versus oracle calls, for a gauge-aware and a gauge-unaware acquisition policy, averaged over seeds with 95% confidence bands.

assumption the replicate validation of Section 5 tests rather than one this section adds. The gauge-zero property is structural: KG$=0$ on gauge directions follows from the kernel inclusion alone, so it does not depend on the weight value, and by Gauss–Markov the value does not improve ranking (Table S11 below). For a *connected* network the inclusion is an equality, $\ker L = \mathrm{span}\{\mathbf{1}\}$, for any positive conductances, and the overall additive offset is then the only unidentifiable direction; a network with k components has a k-dimensional kernel, one offset per component. The Laplacian-precision and resistance-distance machinery is classical generalized least squares (GLS) on graphs and the resistance distance of Klein and Randić [12]; what is specific here is the sandwich precision I_e^2/B_e as the conductance, in place of raw overlap, and the automatic gauge-invariance of the acquisition.

Gauge-aware acquisition (Figure S4). The Fisher–resistance Laplacian (Theorem S1) makes the knowledge gradient exactly zero on gauge (all-ones) contrasts. A controlled simulation gives $\mathrm{KG} = 10^{-21}$ on the all-ones gauge contrast, an exact zero, and the identifiability statement behind it transfers structurally to multi-target networks. A gauge-unaware acquisition spends budget resolving a decision-irrelevant cross-cluster offset, reaching the within-target top-k at regret 0.27 where the gauge-aware one reaches 0.01.

Structural context of the flagged systems (Figure S5). Two of the systems the closure test flags (Figure 7) carry an independently-documented hard structural feature, read here directly from public co-crystal structures. In BACE1 (PDB 4DJW) the catalytic aspartic-acid dyad hydrogen-bonds the inhibitor (Asp93 and Asp289, both 2.7 Å); the protonation state of an aspartic-protease dyad is a recognized source of free-energy error. In BRD4 BD1 (PDB 3MXF, in complex with

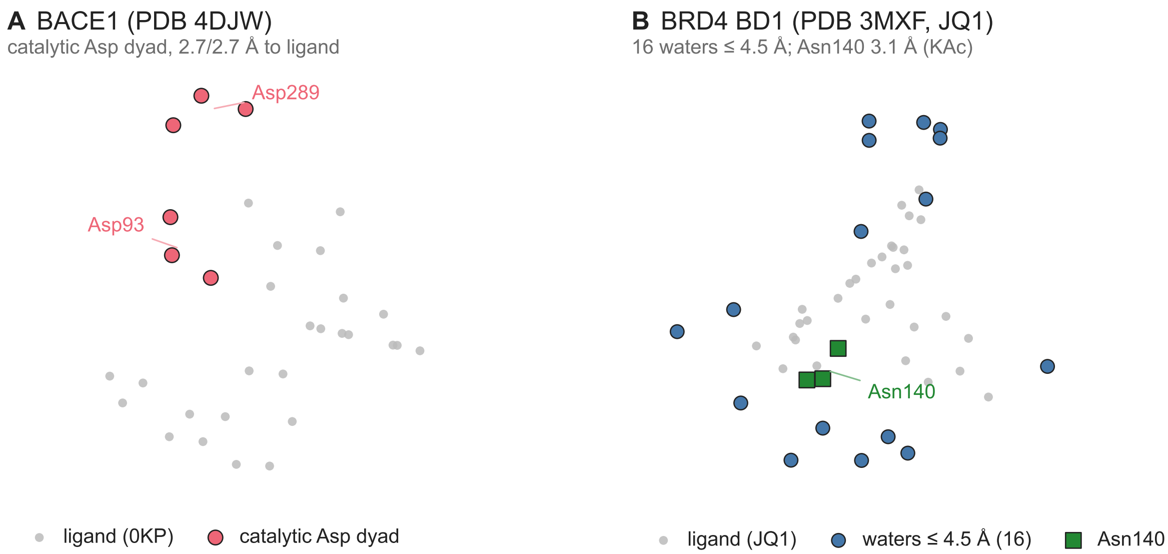

Figure S5: **Structural context of two flagged systems, from public co-crystal structures.** Heavy-atom coordinates. (A) BACE1 (PDB 4DJW): the catalytic aspartic-acid dyad (Asp93/Asp289, red circles) and the bound inhibitor (grey). (B) BRD4 BD1 (PDB 3MXF, in complex with JQ1): crystallographic waters within 4.5 Å of the ligand (blue circles), the recognition residue Asn140 (green squares), and the ligand (grey).

JQ1 [6]) the acetyl-lysine pocket packs 16 crystallographic waters within 4.5 Å of the ligand (8 within 3.5 Å) around the recognition asparagine Asn140 (3.1 Å); this conserved water network is the known BRD4 free-energy challenge [1]. The other flags carry analogous a-priori difficulties: `faah` (covalent serine hydrolase, large acyl channel), `cdk8` and `p38` (large scaffold and R-group changes), and `hif2a` (a buried polar cavity). The context is retrospective and structural rather than a per-edge causal claim: cycle closure is edge-level and blind to node-consistent bias (main text). It establishes only that the flagged set coincides with the features a domain expert would identify a priori. In Figure S5 the colours annotate structural objects rather than methods.

Per-edge observability map and predictive falsifier (Figure S9). The estimation–detection conservation law (Theorem 1) certifies $\sum_e h_e$ = dof to machine precision (maximum deviation 1.78×10^{-14}) across the 48 cyclic systems and identifies 48 structurally un-auditable bridge edges ($h_e = 0$, one per system on average); a pre-registered test of whether the curl-leverage also predicts the location of reproducible systematic error, run on 934 pooled edges over 45 systems against the rule confirmed if and only if $\rho > 0$ and $p < 0.05$ (10^4 within-system block permutations), is negative (studentized Spearman $\rho = -0.073$, $p = 0.9571$), so only the structural certificate is claimed.

Per-system auditability (Table S4). Theorem 2 makes a network's auditable fraction a property of its topology and its reported precisions alone, so it can be tabulated for every benchmark system before any question of what that system's error turns out to be. Table S4 gives, per system, the ligand and edge counts, the number of independent cycles ν, the auditable fraction ν/E, the bridge count, and the median detectable shift over that system's auditable edges. The auditable fraction is exactly $1-(N-c)/E$, for N ligands, E edges and c connected components, and so is small wherever a network is sparse: it runs from 0.12 to 0.46 with a median of 0.31, and no benchmark system exposes more than half of its own error space to the test. The eight flagged systems span essentially that whole range, from 0.12 to 0.46, rather than concentrating at the auditable end: a system is flagged for the error it carries, not for how much of that error is in view. `brd4`, flagged on two of the three replicates, sits at the bottom of the table with a single independent cycle: its reduced χ^2 is a one-degree-of-freedom statistic, which is the quantitative form of the caution Section 6.1 places on it. Rows are ordered by auditable fraction, and the flag marks are recomputed from the three replicates by the same code that draws Figure 2, not transcribed from the main text.

What was pre-registered. Several arguments in this article turn on what was frozen before the computation rather than chosen after it. The design file governing the decomposition increment was committed to the article's repository before the commit that computes any of the quantities it governs, and it is released with the analysis code; the Data and Software Availability statement says where. It fixes three items and no others: H1, the visible fraction of the measured error against experiment with its chance level and its success criterion; H2, the influence-ranked repair race; and H3, the auditability map, which it labels descriptive and gives neither threshold nor verdict. Both success criteria are conjunctive, so a single conjunct firing is not a success. H1 is labelled a quantification backed by an identity rather than a risky test, and says why: experimental affinities are per-ligand, so $\Pi\varepsilon$ reduces to the closure residual and the visible fraction is an identity, with only the comparison against the chance level at risk. The confound the design cannot resolve is declared in it: experimental measurement error and the stereo-blind affinity join are both per-ligand and therefore also gradient fields, so the design cannot separate per-ligand force-field bias from per-ligand experimental or annotation error, and the conclusion it licenses is the first of those and not the second. H2 is declared in the design file to be a second look at data that had already produced one null, and its result carries that status.

The other pre-specified constants this article invokes are recorded in the released analysis code and in the pre-registration for the experiment that fixed them, not in that file: the $|z|$-rule conjunctive criterion, the curl-leverage falsifier rule, the fixed-cutoff head-to-head criterion, the 1.0 kcal/mol cutoff itself, the swept scale grid with its two frozen readouts, the per-edge granularity of the self-calibrated null, and the close-the-loop criterion. Three of them live only in code, and the paragraph below audits those three against the descriptions printed here.

Audit of the frozen criteria that live only in code. Three criteria are recorded in the analysis code and not in the design file, and each is audited here against the description printed in this article. The curl-leverage falsifier is confirmed if and only if $\rho > 0$ with a within-system block-permutation $p < 0.05$ over 10^4 permutations; the code implements that rule and the supplement prints it. The fixed-cutoff head-to-head requires the calibrated test's detection area to exceed both baselines, implemented as a paired bootstrap of $\mathrm{AUC}(A) - \max(\mathrm{AUC}(B), \mathrm{AUC}(C))$ with a win declared only when the point difference is positive and the lower confidence bound excludes zero; the printed description matches, and the tie follows from the $+0.140$ $[-0.008, 0.316]$ against the stronger baseline. The $|z|$-rule conjunctive criterion matches the code as printed here, requiring Stouffer $p < 0.05$ and a guided ΔMUE above the 95th percentile of its own random null in a majority of grounded systems, and that is the upper tail because ΔMUE is defined as MUE(full) − MUE(after), so an improvement is positive. Its frozen file names the lower tail, "below the 5th percentile", which is the same event stated in the opposite sign convention; the code's variable name follows the file rather than the statistic. No result depends on the discrepancy. The frozen file is unaltered, and the description printed in this article is the one the code implements.

The visible fraction is metric-dependent (Section 4). The visible fraction f, the share of the error against experiment lying in the auditable subspace (Section 4), is a ratio of squared norms, so it depends on the metric those norms are taken in, and the main text quotes it in the variance-weighted metric the closure statistic itself uses. Three readings of the same four sub-networks are collected in Table S2, each with its own chance level and with the range the fraction takes when one edge is deleted at a time. The whitened reading takes both quantities in the variance-weighted metric the closure statistic lives in, against dof/E. The isotropic reading keeps that fraction and measures it against $\mathrm{tr}(\Pi W)/\mathrm{tr}(W)$, the chance level an error isotropic in kcal/mol would produce under the same projector; it gives the weakest margins, and it is the reading the abstract and the main text quote. The unweighted reading recomputes the projector itself in the unweighted metric, $\Pi = I - A(A^{\top}A)^{+}A^{\top}$ on the raw incidence matrix A, so neither the fraction nor its chance level carries any variance weighting; there the pooled fraction rises from 0.6% to 5.45%. The worst single cell in the whole table is `bace` under the isotropic reading with one edge deleted, at a factor of 1.36 below its chance level. Both the fraction and the chance level are recomputed on every deletion, since deleting an edge moves the projector as well; `make figHodge` produces the table. `cdk8`'s flag arises from a naming collision between two releases rather than from either release's simulations (Section 6), so the pooled figure is given both ways: 0.0060 whitened and 0.0545 unweighted over the four systems, 0.0060 and 0.0610 with `cdk8` dropped, so the whitened headline does not depend on it. Recomputing the ratio on medians rather than sums, which removes any dependence on a few large residuals, gives 0.0034, 0.0068, 0.0052 and 0.0015, the same order throughout; and `bace`, the one grounded system whose ligand set contains no enantiomer pair collapsed by the stereo-blind affinity join, shows the effect as plainly as the three that do. Every reading puts f below its own chance level; what changes is by how much, and the conclusion the main text draws is the weakest of them.

The label source can also be changed. The measurement uses a stereo-blind, target-wide ChEMBL search under the coverage rule, and the OpenFF protein–ligand benchmark of the downstream audit below (Table S11) carries curated per-edge experimental $\Delta\Delta G$ for three of the four systems (`bace` has none). Matching those to the network by canonical SMILES recovers 57%, 98% and 91% of the curated edges for `cdk8`, `hif2a` and `p38`. The matched sub-networks are much smaller than the grounded ones and `p38`'s retains no cycle, so the check is weak, but where it can be run it agrees: the visible fraction on curated labels is 0.0007 for `cdk8` against a chance level of 0.071, and 0.0006 for `hif2a` against 0.147. Comparing those two figures against what the ChEMBL labels give is only meaningful on identical edges. Against the full systems' ChEMBL fractions, measured over 31 and 59 edges rather than over these 14 and 34, the curated reading is the lower one; restricted to the same edges it is the higher one, 0.00071 and 0.00063 against 0.00016 and 0.00021, by factors of 4.4 and 2.9. The direction on identical edges is the one the decomposition predicts. A per-ligand label error is a gradient field, the whitened projector annihilates it exactly, and it therefore lands entirely in the denominator of f and pushes the fraction down. The stereo-blind, target-wide ChEMBL join is the noisier of the two label sources, and on these same edges its median unsigned error against experiment is the larger of the two, 1.61 against 0.56 kcal/mol on `cdk8` and 1.24 against 1.05 on `hif2a`; the noisier source should therefore give, and does give, the smaller visible fraction. What the two label sets supply is a consistency check on shared calculated values, with the cleaner set giving the larger visible fraction, and not two independent confirmations of either. Lifting the pooled fraction to its chance level by annotation noise alone would require that noise to carry almost all of the total squared standardized error; how much it would have to carry, per system as well as pooled, and how much a literature-scale label noise can actually supply, are measured below (Figure S7). `bace`'s margin is the exception and is why that number does not appear in the abstract.

Table S2: **The visible fraction under three metric conventions.** For each grounded system: edges E, independent cycles ν, the auditable share of that system's error against experiment, the chance level it is measured against, their ratio, the range the fraction takes when one edge is deleted at a time, and the worst margin over those deletions with both the fraction and the chance level recomputed on each. *Whitened* takes both in the variance-weighted metric the closure statistic lives in, against ν/E. *Isotropic* keeps that fraction and measures it against the chance level an error isotropic in kcal/mol would produce under the same projector, $\mathrm{tr}(\Pi W)/\mathrm{tr}(W)$. *Unweighted* recomputes the projector in the unweighted metric, so neither the fraction nor its ν/E chance level carries any variance weighting. Every cell of every convention, and every single-edge deletion, sits below its own chance level; the weakest is `bace` isotropic under deletion. Pooled over the four systems the fraction is 0.0060 whitened and 0.0545 unweighted, and 0.0060 and 0.0610 with `cdk8` dropped. Produced by `make figHodge`.

system	E	ν	metric	visible	chance	margin	leave-one-out	worst
`cdk8`	31	9	whitened	0.0065	0.2903	44.8×	$[0.0034, 0.0113]$	23.60×
			isotropic, kcal/mol	0.0065	0.1273	19.6×	$[0.0034, 0.0113]$	7.36×
			unweighted	0.0286	0.2903	10.2×	$[0.0178, 0.0324]$	8.22×
`hif2a`	59	19	whitened	0.0017	0.3220	187.2×	$[0.0010, 0.0029]$	106.60×
			isotropic, kcal/mol	0.0017	0.0510	29.6×	$[0.0010, 0.0029]$	13.21×
			unweighted	0.0172	0.3220	18.7×	$[0.0127, 0.0188]$	16.53×
`p38`	51	18	whitened	0.0086	0.3529	41.0×	$[0.0033, 0.0132]$	25.81×
			isotropic, kcal/mol	0.0086	0.1240	14.4×	$[0.0033, 0.0132]$	1.46×
			unweighted	0.1371	0.3529	2.6×	$[0.0285, 0.1499]$	2.31×
`bace`	26	5	whitened	0.0460	0.1923	4.2×	$[0.0027, 0.0628]$	2.55×
			isotropic, kcal/mol	0.0460	0.0919	2.0×	$[0.0027, 0.0628]$	1.36×
			unweighted	0.0564	0.1923	3.4×	$[0.0033, 0.0707]$	2.26×

The visible fraction on curated labels, over the benchmark (Figure S6, Table S3). The three-system comparison above runs on three sub-networks the closure test itself selected. The same measurement runs over the whole curated set. Fifteen of the benchmark's targets carry curated per-edge experimental $\Delta\Delta G$, and fourteen of those names also appear in the replicate benchmark (`pde2` does not); after the join, eight of the fourteen have a matched sub-network that still carries a cycle and so a measurement at all. Pooled over those eight, across 205 edges and 31 residual degrees of freedom, the visible fraction is 0.00052, against a dof$/E$ chance level of 0.151 and an isotropic-in-kcal/mol chance level of 0.0137: a factor of 289 below the first and 26 below the second. Split by the flag, the two systems the closure test flags give 0.00071 over 63 edges and the six it does not give 0.00042 over 142. The systems the test does not select show the effect at least as strongly as the ones it does, 35× below the isotropic chance level against 17×, so the measurement is a statement about the benchmark rather than a property of test-selected networks. Every system with a cycle, taken singly, sits one to three orders of magnitude below its own chance level (Table S3). The join rule, the pooling rule and a recovery threshold of 0.50 – recovery being the share of a system's curated edges whose two ligands both resolve into the network – were fixed before any of the eleven systems beyond the three above was measured.

This does not replace the four-system measurement of Section 4 and is not offered as doing so. `bace` is absent from the curated benchmark entirely, `p38`'s matched sub-network is acyclic and contributes nothing, and the expanded set carries 31 residual degrees of freedom against the original's 51. The labels are different labels and the sub-networks are different sub-networks, so the pooled figures are two grounded measurements of different things and neither corrects the other. The expanded pooled figure is 11.5× smaller than the four-system one, and three things change at once to produce that: eleven new systems enter, the matched sub-networks are smaller and sparser in cycles than the full ones, and the label source changes. Only the third is isolated above, on identical edges, and it pushes the other way. What carries across both is the ordering: on every network that carries a cycle, under either label source, the visible fraction sits far below its own chance level.

What the curated join cannot measure, and two ways of keying it (Table S3). Four systems with a real matched sub-network carry no independent cycle in it: `cdk2` ($E = 9$), `cmet` ($E = 2$), `p38` ($E = 16$) and `tyk2` ($E = 10$). A sub-network with no cycle has nothing for a closure diagnostic to see and nothing for the projector to keep, so its visible fraction is identically zero by construction; that is not evidence, and those four contribute to neither side of the pool. Two more join no curated edge at all. `pfkfb3` recovers 2% because the replicate benchmark carries only two `pfkfb3` edges against the curated table's 45, and `thrombin` recovers none because the benchmark's prepared thrombin ligands are protonated where its curated table is neutral and the exact-structure rule refuses the join. Repeating the entire measurement with formal charges stripped from both sides – a rule chosen after seeing that result, and reported as a sensitivity analysis for that reason – gives `thrombin` 14 matched edges. Under the name key those change the pooled figures by nothing at all, because a larger acyclic sub-network is still an acyclic sub-network. Under the structure key used here they do not: merged, they carry two cycles and a visible fraction of 0.115, and the pooled figure moves to 0.00445. That is an artefact of the key. The releases name one molecule in several poses, `3b` in the JACS set against `3b pose 1` in the water set, so a 2D structure merges nodes a release deliberately keeps apart and the cycles it creates assert that two poses of two preparations have one free energy. It is the mirror of the `cdk8` collision, and the reason a name not being an identifier does not make a 2D structure one. The measurement never meets it: the frozen exact rule refuses every `thrombin` join on protonation state, so the system contributes nothing to the figures reported above. The declared recovery threshold likewise changes nothing here: the two systems it excludes already contribute no matched edge. Both readings are reported; neither replaces the other. Restricting a network to the edges a curated table happens to cover also removes cycles: the pooled dof$/E$ here is 0.151 against the whole benchmark's 0.325, which is why both chance levels are carried in every row rather than one.

Two keying decisions move the numbers. The first is how a network ligand name is resolved to a structure. `cdk8`'s two releases reuse seven ligand names for chemically different molecules; resolving each edge through the prepared input structures of its own release, rather than through one name-to-structure table per system in which the second release silently overwrites the first, recovers 44 of 46 curated edges rather than 57% of them and gives a 29-edge matched sub-network rather than a 15-edge one. `cdk8`'s visible fraction rises from 0.0007 to 0.0023 as a result, against chance levels of 0.172 and 0.077. The release-local rule is the less favourable of the two for the argument here and is the one used throughout. The second decision was undeclared in the protocol above, which fixes how edges are matched but not how the matched sub-network's nodes are identified, and the same finding settles it. Keying nodes by ligand name gives `ptp1b` $f = 0.0023$; keying them by resolved structure, which merges two names in different releases that denote the same molecule and knits two components into one, gives 0.0159, and the worst margin any single-edge deletion leaves under the isotropic chance level falls from 10.3× to 2.0×. `ptp1b` is the only system of the eight whose matched sub-network changes at all under the alternative key. The figures quoted here and in Table S3 are the structure-keyed ones, which are also the less favourable of the two: merging nodes adds cycles, so the pooled fraction rises from 0.00046 to 0.00052 and the not-flagged pool from 0.00032 to 0.00042. Both readings are generated by `make figGround` and recorded in full.

The label-noise floor under the visible fraction (Figure S7). Experimental and annotation error is a property of a ligand, so it enters a per-edge error as a difference of per-ligand terms, $\epsilon_e \to \epsilon_e + b_j - b_i$: a gradient field, which the whitened residual projector annihilates. Injecting such a field at a per-ligand σ of 0.93 kcal/mol into the four grounded sub-networks, 200 seeded draws each, moves the numerator $\|\Pi\tilde{\epsilon}\|^2$ by at most 1.96×10^{-11} of its own size over all 800 draws, against a tolerance of 10^{-10} fixed before the measurement was run, and by at most 1.3×10^{-13} measured against the injected field's own scale; the denominator meanwhile grows by 1.6 to 2.2× at the median. Label noise therefore contributes nothing to the numerator of f and all of it to the denominator, and any correction for it can only move f upward. The 0.6% the main text reports is a floor on the share of the error attributable to the calculation's own bias, not an estimate of it.

How much room that leaves is exact arithmetic rather than a fit. If a share r of the measured per-edge error variance is label noise, the denominator shrinks by $s = 1/(1-r)$ and f rises to sf, the numerator being fixed. Pooled, f would first reach its chance level at a shrinkage of 50.6× in the whitened metric and 14.1× in the isotropic one. That pooled crossing is not the figure to quote on its own, because per system the room is far from uniform: `bace` needs 4.2× whitened and 2.0× isotropic, and under the least favourable single-edge deletion `p38` needs 1.46× isotropic and `bace` 1.36×. Against those, three declared levels of label noise say what a correction could supply. The first, L1, is the measured literature quantity: mixed public affinity data reproduce across laboratories at 0.68 $\log_{10}$ units per ligand [11], which at 1.3642 kcal/mol per $\log_{10}$ unit is 0.93 kcal/mol per ligand and 1.31 kcal/mol per edge. L2 and L3 carry half and a quarter of L1's per-edge variance and are stated reductions of it rather than independent measurements.

The verdict differs by metric, and the metric it fails in is the one the abstract and the main text quote. In the isotropic metric the bound does not survive a label-noise correction uniformly. `bace` isotropic needs 2.00× and L2 supplies 2.55×; `bace` isotropic under the least favourable single-

Table S3: **The visible fraction on curated labels, over the benchmark.** For each target whose curated per-edge experimental $\Delta\Delta G$ the replicate benchmark can be joined to: curated edges E_{cur}, the share of them whose two ligands both recover into the network (ρ), and the matched sub-network's ligands N, edges E, components c and independent cycles dof. Then the visible fraction f of the error against those labels, its dof$/E$ chance level, the chance level $\text{tr}(\Pi W)/\text{tr}(W)$ an error isotropic in kcal/mol would produce under the same projector, the range of f over single-edge deletions, the worst margin chance$/f$ any deletion leaves under each chance level, and the median $|\epsilon|$ against experiment. Both the fraction and its chance level are recomputed on every deletion, since deleting an edge moves the projector as well. f and its deletion range are given in units of 10^{-3}. A dagger marks the systems the closure test flags. The systems whose matched sub-network carries no cycle, and so no measurement, are named below the rule.

system	E_{cur}	ρ	N	E	c	dof	$f/10^{-3}$	dof$/E$	iso	f range$/10^{-3}$	worst margin		$\lvert\epsilon\rvert$	
											dof$/E$	iso		
`cdk8`†	46	0.96	29	29	5	5	2.28	0.172	0.077	$[1.59, 2.64]$	54×	6.2×	1.08	
`eg5`	43	1.00	26	32	3	9	2.07	0.281	0.095	$[1.04, 4.75]$	54×	12.0×	0.92	
`hif2a`†	51	0.98	33	34	4	5	0.63	0.147	0.005	$[0.28, 1.00]$	151×	5.4×	1.05	
`mcl1`	36	1.00	23	21	3	1	0.26	0.048	0.007	$[0.26, 0.40]$	124×	22.9×	0.60	
`ptp1b`	36	0.94	17	19	2	4	15.86	0.211	0.061	$[2.32, 19.25]$	9×	2.0×	1.15	No
`shp2`	37	1.00	16	12	5	1	0.08	0.083	0.009	$[0.08, 0.24]$	372×	42.5×	1.02	
`syk`	66	0.97	39	35	8	4	0.38	0.114	0.007	$[0.10, 0.70]$	168×	12.0×	0.59	
`tnks2`	36	0.86	25	23	4	2	0.03	0.087	0.003	$[0.00, 0.05]$	1558×	22.8×	0.64	
pooled, all (8 systems)				205		31	0.52	0.151	0.014					
pooled, flagged (2 systems)				63		10	0.71	0.159	0.012					
pooled, not flagged (6 systems)				142		21	0.42	0.148	0.015					

measurement, matched sub-network acyclic: `cdk2` ($E = 9$, dof $= 0$), `cmet` ($E = 2$, dof $= 0$), `p38` ($E = 16$, dof $= 0$), `tyk2` ($E = 10$, dof $= 0$). No measurement, no curated edge joined: `pfkfb3` ($\rho = 0.02$), `thrombin` ($\rho = 0.00$).

edge deletion needs 1.36×, which both L2 and L3 supply; `p38` isotropic under one deletion needs 1.46× and L3, a quarter of the mixed-laboratory variance, supplies 1.80×. A plausible label-noise correction therefore erases the isotropic reading on those two systems. In the whitened metric the pooled bound needs a 50.6× shrinkage to be lost and the most any level supplies pooled is 4.13× once each system is held to its own arithmetic, so there it survives by an order of magnitude; the single exception is `bace` under the least favourable single-edge deletion, where the 2.55× needed and the 2.55× L2 supplies coincide to within one per cent. The bound is therefore robust to a label-noise correction in the whitened metric the fraction is defined in, and is not robust in the isotropic metric on `p38` and `bace` – the two systems whose margins were the weakest before any correction was considered.

The isotropic readings just given stand: L2 and L3 erase them, and nothing here refutes those two levels. These data do refute L1, the only level with a literature measurement behind it. A mixed-laboratory per-ligand σ of 0.68 $\log_{10}$ implies a label-noise share $r \geq 1$ on `cdk8`, `p38` and `bace`: more label variance than each of those networks' entire measured error against experiment contains, which is arithmetically impossible, and is reported as impossible and not clipped. That is direct evidence, from the data and not from an assumption imported to protect the bound, that 0.68 $\log_{10}$ is an upper bound and not an estimate for this article's grounding, which takes one ChEMBL target and one assay type per system and never mixes IC_{50} with K_i.

Two numbers in the same arithmetic land close together without being the same quantity. The 14.13× that L1 supplies pooled without the per-system cap sits within one per cent of the 14.08× needed to reach the pooled isotropic chance level. They are different quantities, and L1 is in any case impossible on three of the four systems taken singly.

None of this measures the label noise in these particular ChEMBL series; it bounds what a literature-scale label noise could do. A direct measurement would need replicate assay values per ligand, which the public sources do not carry.

The system column carries the benchmark's own name for a target's edge network, which is not a protein: the 49 names span 41 protein labels, `bace` contributing four and `hsp90` four, so a count of flagged systems is not a count of flagged targets. Nor is a name always one released data set: eight of them combine two or more, which the cross-release paragraph below takes up. `ciordia_retro` is released under the `asap_mix` group, and the release's own statement is the only target assignment used for it.

The component column records the number of connected pieces of a system's edge network. Six of the 48 systems are disconnected (`cdk2`, `jnk1`, `mcl1`, `p38`, `ptp1b`, `thrombin`), each into two pieces, so the components sum to 54: the benchmark's edge lists join congeneric series that share a target without joining them to each other. Nothing in the closure statistic is affected, since dof is computed as E minus the incidence rank throughout and that rank already accounts for the extra kernel direction. Elsewhere disconnection does matter. The offset that Theorem 2(iv) says the first anchor removes is one offset per component, so a disconnected system needs one anchor in each piece before any further anchor resolves a dimension. The grounded networks of Section 4 inherit the same structure, `cdk8` and `bace` being disconnected once restricted to ligands carrying affinities. Their mean unsigned error against experiment is reported after a single global mean alignment, which is the conservative convention: aligning each component to its own experimental mean instead, which uses more experimental information than a prediction has access to, lowers the error by 0.067 and 0.014 kcal/mol respectively and leaves the two connected systems unchanged.

The cost of a network that can be audited, per system (Table S5). The design rule of Section 3 is the prospective counterpart of the table above, and Table S5 gives it per system. Two

Table S4: **What each benchmark system can be audited for.** For the 48 systems carrying at least one independent cycle: N ligands, E edges, c connected components, ν independent cycles, the auditable fraction ν/E of that system's error space, the number of bridge edges (br), which lie in no cycle and carry no evidence at any magnitude, and the median shift $\tilde{\delta}^*$ in kcal/mol at which the closure statistic carries unit noncentrality, over that system's auditable edges. Rows are ordered by auditable fraction ascending. A dagger marks the eight systems the Benjamini–Hochberg test flags on at least one of the three replicates. Values are replicate-0 and are produced by the same run of the released analysis code that draws Figure 2. Note that $\tilde{\delta}^*$ is not a detection threshold: reaching 80% power at $\alpha = 0.05$ needs a shift of 2.8 to 4.7 times it, and it is carried on the raw reported standard errors.

system	N	E	c	ν	ν/E	br	$\tilde{\delta}^*$	system	N	E	c	ν	ν/E	br	$\tilde{\delta}^*$
`brd4`†	8	8	1	1	0.12	0	0.49	`mcl1`	54	76	2	24	0.32	1	0.91
`taf12`	8	8	1	1	0.12	0	0.48	`hif2a`†	41	59	1	19	0.32	1	0.78
`scyt_dehyd`	7	7	1	1	0.14	0	0.43	`shp2`	26	37	1	12	0.32	2	1.14
`chk1`	13	15	1	3	0.20	1	1.29	`syk`	46	67	1	22	0.33	5	1.25
`t4_lysozyme`	12	14	1	3	0.21	0	0.39	`bace1`	3	3	1	1	0.33	0	0.64
`hsp90_kung`	11	13	1	3	0.23	0	0.56	`dlk`	5	6	1	2	0.33	0	2.09
`liga`	11	13	1	3	0.23	1	0.95	`factor_xa`	3	3	1	1	0.33	0	2.46
`jnk1`	24	29	2	7	0.24	3	0.93	`ptp1b`	26	36	2	12	0.33	5	1.39
`ephx2`	4	4	1	1	0.25	0	3.85	`renin`†	29	42	1	14	0.33	0	0.64
`hsp90_single_ring`	7	8	1	2	0.25	0	0.53	`thrombin`	37	54	2	19	0.35	0	0.55
`hsp90_woodhead`	4	4	1	1	0.25	1	0.37	`keranen_p2`	12	17	1	6	0.35	0	1.27
`irak4_s3`	4	4	1	1	0.25	1	1.60	`jak2_set1`	10	14	1	5	0.36	1	0.74
`urokinase`	4	4	1	1	0.25	0	0.41	`p38`†	40	60	2	22	0.37	5	1.02
`tnks2`	27	35	1	9	0.26	0	0.47	`hiv1_protease`	13	19	1	7	0.37	0	0.46
`faah`†	24	31	1	8	0.26	3	0.88	`cdk2`	19	27	2	10	0.37	0	0.92
`bace_ciordia_prospective`	9	11	1	3	0.27	0	0.68	`eg5`	28	43	1	16	0.37	3	1.51
`bace`†	36	49	1	14	0.29	2	0.57	`btk`	6	8	1	3	0.38	0	1.63
`bace_p3_arg368_in`	21	28	1	8	0.29	8	0.71	`tyk2`	19	29	1	11	0.38	1	0.68
`hsp90_2rings`	6	7	1	2	0.29	0	0.13	`itk`	4	5	1	2	0.40	0	1.24
`jak1`	6	7	1	2	0.29	1	1.60	`cmet`	24	39	1	16	0.41	0	1.12
`mup1`	6	7	1	2	0.29	0	0.28	`jak2_set2`	8	12	1	5	0.42	0	1.71
`hne`	17	23	1	7	0.30	0	1.25	`egfr`	5	7	1	3	0.43	0	2.90
`galectin`	26	36	1	11	0.31	0	0.61	`irak4_s2`	5	7	1	3	0.43	0	0.98
`ciordia_retro`†	32	45	1	14	0.31	0	0.68	`cdk8`†	35	63	1	29	0.46	3	0.98

kinds of number sit in it and the table keeps them apart. The augmentation counts a_b and a_2, 29 and 36 added ligand pairs over the benchmark, are pure topology: they assume nothing whatever about the edges they add, are certified minimum against the Eswaran–Tarjan augmentation bound, and were checked independently by brute force on 39 of the 40 (system, target) pairs that need any edge at all, every subset of one fewer edge failing on each. The further columns, the edges a greedy design adds to bring a system's median δ^* below 1.0, 0.75 and 0.5 kcal/mol, assume each new edge carries that system's median reported variance, and are an achievable cost rather than a proven minimum. That assumption is what the third column of the sensitivity check tests: sweeping the assumed variance by a factor of two in each direction leaves the verdict and the count stable to within one edge on 93 of 96 system-by-scale comparisons. Two consequences belong with the rule. Removing bridges buys coverage and not resolution: on three of the 21 systems that need any augmentation the median δ^* taken over all edges is *worse* after augmentation than the median over auditable edges alone that Table S4 reports, because the bridges that median silently excludes are now inside it. The second is that $h_e \leq 1$ pointwise gives $\delta_e^* \geq \mathrm{se}_e$, so no topology can push an edge below its own reported standard error; on a complete graph over N ligands $h_e = 1 - 2/N$, a second floor for small systems. Measuring every remaining ligand pair still leaves nine systems above 1.0 kcal/mol and nineteen above 0.5, which is the bound of Theorem 2 appearing in a design question rather than in a diagnostic one.

Between-preparation variance: what the multi-release systems can and cannot supply (Figure S8). Three passages of this article need the same missing quantity: the variance between independent *preparations* of one target, relative to the sampling variance the reported bar carries. The three are the disagreement with Wade et al. [19] over whether that bar is conservative, the observation that nothing flags once every bar is widened by half, and the caveat that the three repeats share one starting structure. The benchmark appears to offer it in one place. Eight system names union edges from more than one separately released data set, and where two releases share ligands the union carries cycles that exist in neither release alone, which close only if the two preparations agree; splitting the closure statistic into its within-release and across-release blocks would then measure the excess directly. That split, run over every multi-release system and all three repeats, shows that the public data do not supply the quantity. Two shared ligands are needed before the union carries an across-release cycle, and only two of the eight reach that: in the other six (`cdk2`, `jnk1`, `mcl1`, `p38`, `ptp1b`, `thrombin`) the releases share at most one ligand, the union is disconnected or joined by a tree, and no measurement exists at any magnitude.

Of the two that remain, one is not a join between preparations at all. `cdk8`'s across-release block is large, at a reduced χ^2 of 7.77, 4.79 and 8.41 over the three repeats against a null of exactly 1, and 5.8 to 8.2 times its own within-release block. That excess is the naming collision of Section 6 and not a disagreement between two preparations: its two releases reuse seven ligand names for chemically different molecules, so every spanning cycle asserts an equality between distinct ligands. Disambiguating by release removes all six across-release cycles on every repeat and takes the system's degrees of freedom from 29 to the within-release total of 23, which is the correction this article already applies. The across-release excess must not be read as a between-preparation estimate, and it is not used as one anywhere here.

That leaves `tyk2` as the only genuine cross-release join in the benchmark: two shared ligands, one across-release cycle. Its across-release block sits at a reduced χ^2 of 0.18, 0.70 and 0.04 over the three repeats, $p = 0.67$ at the first, with a ratio to the within-release block of 0.2, 0.6 and 0.0. The null it is read against is the right one: synthetic draws on each system's real topology and real reported bars, with one node potential per shared ligand and independent noise of the

Table S5: **The cost, in added edges, of a network that can be audited.** For the 48 benchmark systems carrying at least one independent cycle: E edges and the number of bridge edges (br), then two certified-minimum augmentation counts. a_b is the smallest number of new ligand pairs that leaves NO bridge, so that every edge lies on a cycle and carries some evidence; it treats a system's connected components separately, which is all auditability requires, since the closure fit already absorbs one offset per component. a_2 additionally joins those components, leaving the whole network 2-edge-connected. Both are pure topology and assume nothing about the new edges. The last three columns are the FURTHER edges a greedy design adds, on top of a_2, to bring the median δ^* over that system's own E edges below 1.0, 0.75 and 0.5 kcal/mol, each new edge assumed to carry that system's median reported variance; unlike a_b and a_2 these are an achievable cost and not a proven minimum, since the design is greedy. - marks a target no topology can reach, because even measuring every remaining ligand pair leaves the median above it, and $>2E$ a target not reached within the explored budget. Across the benchmark 29 added edges remove all 48 bridges and 36 leave every system 2-edge-connected, 2.5% and 3.1% of the 1143 edges already run. δ^* is the shift at unit noncentrality and not a detection threshold; 80% power at $\alpha = 0.05$ needs 2.8 to 4.7 times it, so these are targets on a resolution scale. Values are replicate-0 and come from the same run of the released analysis code, `make figDesign`. A dagger marks the six systems the cycle-closure test flags.

system	E	br	a_b	a_2	+1.0	+0.75	+0.5	system	E	br	a_b	a_2	+1.0	+0.75	+0.5
p38†	60	5	3	4	1	13	–	ciordia_retro	45	0	0	0	0	0	4
bace_p3_arg368_in	28	8	3	3	1	1	12	galectin	36	0	0	0	0	0	2
ptp1b	36	5	2	3	62	–	–	hiv1_protease	19	0	0	0	0	0	0
cdk2	27	0	0	2	0	2	27	hsp90_2rings	7	0	0	0	0	0	0
cdk8†	63	3	2	2	0	45	–	hsp90_kung	13	0	0	0	0	0	1
faah†	31	3	2	2	0	1	8	hsp90_single_ring	8	0	0	0	0	0	1
jnk1	29	3	2	2	0	2	9	irak4_s2	7	0	0	0	0	–	–
mcl1	76	1	1	2	0	1	3	mup1	7	0	0	0	0	0	0
thrombin	54	0	0	2	0	0	1	renin	42	0	0	0	0	0	2
syk	67	5	2	2	2	5	58	scyt_dehyd	7	0	0	0	0	0	0
eg5	43	3	2	2	–	–	–	t4_lysozyme	14	0	0	0	0	0	0
bace†	49	2	1	1	0	0	1	taf12	8	0	0	0	0	0	0
hif2a†	59	1	1	1	0	1	12	tnks2	35	0	0	0	0	0	0
hsp90_woodhead	4	1	1	1	0	0	0	urokinase	4	0	0	0	0	0	0
jak2_set1	14	1	1	1	0	0	2	cmet	39	0	0	0	1	16	–
liga	13	1	1	1	0	0	4	keranen_p2	17	0	0	0	1	3	–
tyk2	29	1	1	1	0	0	2	btk	8	0	0	0	2	–	–
chk1	15	1	1	1	2	16	24	hne	23	0	0	0	3	$>2E$	–
shp2	37	2	1	1	2	7	–	dlk	6	0	0	0	–	–	–
irak4_s3	4	1	1	1	–	–	–	egfr	7	0	0	0	–	–	–
jak1	7	1	1	1	–	–	–	ephx2	4	0	0	0	–	–	–
bace1	3	0	0	0	0	0	–	factor_xa	3	0	0	0	–	–	–
bace_ciordia_prospective	11	0	0	0	0	0	3	itk	5	0	0	0	–	–	–
brd4†	8	0	0	0	0	0	0	jak2_set2	12	0	0	0	–	–	–

reported size, put `tyk2`'s block at a mean reduced χ^2 of 1.069 over 2000 draws per repeat, rejecting at 5.7 to 5.8 per cent, and `cdk8`'s at 1.001. A second preparation of `tyk2` therefore adds nothing detectable beyond sampling, on one cycle, which is far too little to estimate a variance from. The public data contain no usable estimate of between-preparation variance, and the disagreement with Wade et al. [19] and the widened-bar observation are scoped by that absence rather than settled by it. Four further limits apply to any future reading of this split and none is removable from these data: an across-release block confounds preparation with force field, protonation, starting structure and mapping, so it bounds the preparation term from above rather than isolating it; two separately released data sets are not two replicates of one protocol; the counts are small enough that a single edge moves them, which the leave-one-edge-out ranges emitted with the figure quantify; and the split is not independent of what this article already reports, since the one system carrying most of the arithmetic is one the closure test already flags for the reason given above.

Close-the-loop accuracy comparison (Figure S10). On the four flagged systems groundable against public ChEMBL affinity, guided removal of the QC-flagged high-$|z|$ edges ties a size-matched random removal on MUE-vs-experiment (combined Stouffer $p = 0.48$). Removing the highest-$|z|$ edges lowers the closure statistic (Figure 3A), but that is arithmetic and not evidence, because the statistic is the sum of exactly the per-edge quantities being removed in descending order (Section 6). The comparison adds that the same removal yields no improvement against experiment. The reach demonstrated for the closure test therefore stops at internal edge-level consistency and does not extend to force-field-limited absolute accuracy, the same scope boundary as the clean `eg5` system. Throughout, ΔMUE is oriented as MUE(full) − MUE(after), the same orientation as Figure 2C.

Fixed-cutoff head-to-head and χ^2 reconciliation (Figure S11). Two pre-registered checks were run on the same 1143 edges of the 48 cyclic systems. The first is the head-to-head: the calibrated test, scored by area under the receiver-operating-characteristic curve (AUC) against the reproducible-systematic magnitude defined in Computational details, scores 0.865. Against a fixed 1.0 kcal/mol hysteresis cutoff's 0.575 the paired bootstrap difference is $+0.290$ $[+0.094, +0.490]$; against a fixed-se χ^2 test's 0.725 it is $+0.140$ $[-0.008, +0.316]$, straddling zero. The pre-registered criterion required the calibrated test to exceed both, so the verdict is a tie. The ordering within that tie is not a discrimination gain over the fixed cutoff in standard use: the scored label standardizes each residual by the same per-edge reported variance the calibrated rule weights by, so the AUC ordering is monotone in a rule's proximity to that variance and the comparison ranks nothing. The second check tested whether correlation among residuals of edges sharing a ligand endpoint explains the apparent gap between the inverted-median factor, 1.71× obtained as $1/\sqrt{0.34}$ from the median reduced χ^2, and the replicate ratio of 1.41× measured against seed and velocity repeats. It does not: the excess correlation is -0.0079 $[-0.0314, +0.0164]$, and the Benjamini–Hochberg flag set is unchanged.

The two inverted-median factors quoted in this article are the same functional on two populations and do not conflict. The pre-registered constant 1.71× is $1/\sqrt{0.34}$ from the median reduced χ^2 over all 48 cyclic systems, the population of Figure 7; the 1.67× annotated inside panel B is $1/\sqrt{0.361}$ over the 46 systems this second check admits, since estimating the shared-node null needs at least one disjoint edge pair per system and `bace1` and `factor_xa` are three-edge triangles that have none. Dropping those two raises the median.

Neither the pre-registered constant nor the factor annotated in panel B is a conservatism estimate, because the premise of the second check is unsound: by Theorem 1 the closure statistic identifies the closure-weighted mean of c_e^{-2} and not a median, so $1/\sqrt{\text{median}}$ does not estimate the same functional as the replicate ratio. There is accordingly no physical gap for shared-node correlation to explain. The pre-registered constants are retained as the pre-registered record, and the verdict above is reported as run and not revised. On replicate 0 the calibrated test flags 6 of 48 systems, the fixed-se χ^2 test 11, and the fixed cutoff 26, and the three flag sets are strictly nested (calibrated $\subset$ fixed-se $\subset$ fixed cutoff), so on that replicate the calibrated test catches no system the fixed cutoff misses and instead removes 20 of its flags. The strict chain is a property of that replicate and not of the benchmark. On replicate 1 the counts are 6, 13 and 24 and two links fail: `ciordia_retro` is flagged by the calibrated test and not by the fixed-se test, and `mup1` by the fixed-se test and not by the cutoff. On replicate 2 the counts are 3, 11 and 24 and the chain holds. The calibrated set lies inside the cutoff's on all three, which is the only containment any claim here rests on. Under the release's own blocks rather than the by-target grouping, the same replicate-0 test over 56 testable blocks gives 6, 10 and 29, the chain nests, and the calibrated test removes 23 of the cutoff's flags. Whether that removal is an improvement is what the head-to-head above cannot determine, for the reason given there. The 1.0 kcal/mol cutoff was pre-registered at the field-standard value before this comparison was run, and no sweep over alternative thresholds was performed, so the comparison above is specific to that one fixed rule.

The flag sets under arbitrary dependence (Figure S12). The correction reported throughout is Benjamini–Hochberg, whose guarantee needs a dependence assumption between systems. Benjamini–Yekutieli [5] needs none, at the cost of running the same step-up rule at α/H_m with $H_{48} = 4.4588$, that is at 0.0112 rather than 0.05. It returns {`bace`, `brd4`, `cdk8`, `faah`, `hif2a`, `p38`} on replicate 0, identical to the published set; {`bace`, `cdk8`, `renin`} on replicate 2, again identical; and on replicate 1 {`bace`, `ciordia_retro`, `hif2a`, `p38`, `renin`}, which drops `brd4` from the Benjamini–Hochberg set of six. The published set therefore does not depend on the dependence assumption, and the variation of the flag set across replicates is unchanged by removing it. `make figStab` emits the three sets.

Per-system test statistics behind the flag sets (Tables S6 and S7). The flag sets summarized in Figure S12 and in Section 6.1 are thresholded Benjamini–Hochberg adjusted p-values, so the underlying continuous quantities are reported in full rather than only their dichotomization. For each system, a generalized-least-squares cycle-closure fit under the calibrated per-edge null gives a χ^2 on ν degrees of freedom, ν being the number of independent cycles in that system's edge network; the reduced value $\chi^2_\nu = \chi^2/\nu$ and the raw upper-tail probability follow, and those raw p-values are corrected for testing 48 systems by the Benjamini–Hochberg procedure at a false-discovery rate of 0.05, giving the adjusted q-values tabulated below. The $q \leq 0.05$ rule reproduces the released implementation exactly, and that correspondence is asserted on every set the analysis emits. Each of the benchmark's three protocol replicates is analysed independently: the fit is rerun from scratch on that replicate's edges and the correction is applied within that replicate alone, so the three q columns are not a joint error rate across replicates. All values in both parts of the table are produced by the same run of the released analysis code that draws Figure S12.

Out-of-sample behaviour of the calibrated null (Figure S13). Each candidate held-out statistic is calibrated against a no-error null before use, and the realized level of every candidate is reported. Two of the candidates fire far above nominal and are not used; the remaining two hold their nominal level and are the ones carried into the selection comparisons.

Table S6: **Per-system cycle-closure statistics and Benjamini–Hochberg adjusted q-values, per replicate (part 1 of 2).** Reduced χ^2 and adjusted q-value from the calibrated cycle-closure test at a false-discovery rate of 0.05, computed independently on each of the three protocol replicates, for the 48 systems with at least one independent cycle; ν is the number of independent cycles. Bold marks $q \leq 0.05$; the final column lists the replicates on which the system is flagged, and a dash where it never is. Edge and cycle counts are the replicate-0 values. Rows are ordered by replicate-0 q ascending. The remaining 24 systems continue in Table S7.

			replicate 0		replicate 1		replicate 2		
system	edges	ν	χ^2_ν	q	χ^2_ν	q	χ^2_ν	q	flagged
bace	49	14	**5.57**	$\mathbf{3.2{\times}10^{-9}}$	**7.33**	$\mathbf{7.0{\times}10^{-14}}$	**7.10**	$\mathbf{2.9{\times}10^{-13}}$	0, 1, 2
cdk8	63	29	**2.68**	$\mathbf{6.1{\times}10^{-5}}$	1.45	$2.8{\times}10^{-1}$	**2.62**	$\mathbf{1.1{\times}10^{-4}}$	0, 2
brd4	8	1	**15.40**	$\mathbf{1.4{\times}10^{-3}}$	**8.05**	$\mathbf{3.6{\times}10^{-2}}$	4.94	$1.7{\times}10^{-1}$	0, 1
faah	31	8	**3.71**	$\mathbf{2.0{\times}10^{-3}}$	0.45	$1.0{\times}10^{0}$	1.47	$7.1{\times}10^{-1}$	0
hif2a	59	19	**2.53**	$\mathbf{2.0{\times}10^{-3}}$	**2.96**	$\mathbf{2.4{\times}10^{-4}}$	2.05	$5.4{\times}10^{-2}$	0, 1
p38	60	22	**2.40**	$\mathbf{2.0{\times}10^{-3}}$	**2.48**	$\mathbf{1.7{\times}10^{-3}}$	1.81	$1.1{\times}10^{-1}$	0, 1
mcl1	76	24	1.82	$5.5{\times}10^{-2}$	1.79	$6.8{\times}10^{-2}$	1.04	$1.0{\times}10^{0}$	–
bace_p3_arg368_in	28	8	1.88	$3.5{\times}10^{-1}$	1.88	$2.8{\times}10^{-1}$	2.32	$1.4{\times}10^{-1}$	–
taf12	8	1	3.21	$3.9{\times}10^{-1}$	0.15	$1.0{\times}10^{0}$	0.33	$1.0{\times}10^{0}$	–
liga	13	3	2.10	$4.7{\times}10^{-1}$	0.76	$1.0{\times}10^{0}$	1.80	$6.9{\times}10^{-1}$	–
thrombin	54	19	1.30	$7.4{\times}10^{-1}$	0.86	$1.0{\times}10^{0}$	0.61	$1.0{\times}10^{0}$	–
renin	42	14	1.29	$8.1{\times}10^{-1}$	**2.81**	$\mathbf{3.1{\times}10^{-3}}$	**3.32**	$\mathbf{3.8{\times}10^{-4}}$	1, 2
ciordia_retro	45	14	0.89	$1.0{\times}10^{0}$	**3.86**	$\mathbf{3.1{\times}10^{-5}}$	0.40	$1.0{\times}10^{0}$	1
tyk2	29	11	0.79	$1.0{\times}10^{0}$	1.06	$1.0{\times}10^{0}$	0.78	$1.0{\times}10^{0}$	–
ephx2	4	1	0.77	$1.0{\times}10^{0}$	1.67	$7.7{\times}10^{-1}$	1.33	$9.3{\times}10^{-1}$	–
shp2	37	12	0.74	$1.0{\times}10^{0}$	1.21	$8.7{\times}10^{-1}$	0.64	$1.0{\times}10^{0}$	–
itk	5	2	0.68	$1.0{\times}10^{0}$	0.08	$1.0{\times}10^{0}$	0.10	$1.0{\times}10^{0}$	–
tnks2	35	9	0.63	$1.0{\times}10^{0}$	0.43	$1.0{\times}10^{0}$	2.07	$1.7{\times}10^{-1}$	–
mup1	7	2	0.55	$1.0{\times}10^{0}$	3.05	$2.8{\times}10^{-1}$	0.29	$1.0{\times}10^{0}$	–
scyt_dehyd	7	1	0.54	$1.0{\times}10^{0}$	1.34	$8.5{\times}10^{-1}$	1.23	$9.3{\times}10^{-1}$	–
cdk2	27	10	0.48	$1.0{\times}10^{0}$	1.33	$7.7{\times}10^{-1}$	1.22	$9.3{\times}10^{-1}$	–
eg5	43	16	0.48	$1.0{\times}10^{0}$	0.40	$1.0{\times}10^{0}$	0.30	$1.0{\times}10^{0}$	–
galectin	36	11	0.37	$1.0{\times}10^{0}$	0.22	$1.0{\times}10^{0}$	0.88	$1.0{\times}10^{0}$	–
jak2_set1	14	5	0.35	$1.0{\times}10^{0}$	0.83	$1.0{\times}10^{0}$	0.15	$1.0{\times}10^{0}$	–

Table S7: **Per-system cycle-closure statistics and Benjamini–Hochberg adjusted q-values, per replicate (part 2 of 2).** Table S6 continued: the remaining 24 of the 48 systems, in the same order and with the same conventions. No system in this part is flagged on any replicate.

			replicate 0		replicate 1		replicate 2		
system	edges	ν	χ^2_ν	q	χ^2_ν	q	χ^2_ν	q	flagged
`hsp90_2rings`	7	2	0.34	$1.0{\times}10^0$	0.29	$1.0{\times}10^0$	0.12	$1.0{\times}10^0$	–
`factor_xa`	3	1	0.31	$1.0{\times}10^0$	0.01	$1.0{\times}10^0$	0.02	$1.0{\times}10^0$	–
`keranen_p2`	17	6	0.29	$1.0{\times}10^0$	0.19	$1.0{\times}10^0$	0.04	$1.0{\times}10^0$	–
`chk1`	15	3	0.28	$1.0{\times}10^0$	0.27	$1.0{\times}10^0$	0.69	$1.0{\times}10^0$	–
`hiv1_protease`	19	7	0.28	$1.0{\times}10^0$	0.43	$1.0{\times}10^0$	0.94	$1.0{\times}10^0$	–
`irak4_s2`	7	3	0.25	$1.0{\times}10^0$	0.15	$1.0{\times}10^0$	0.04	$1.0{\times}10^0$	–
`t4_lysozyme`	14	3	0.25	$1.0{\times}10^0$	0.57	$1.0{\times}10^0$	2.14	$5.0{\times}10^{-1}$	–
`ptp1b`	36	12	0.21	$1.0{\times}10^0$	0.20	$1.0{\times}10^0$	0.58	$1.0{\times}10^0$	–
`btk`	8	3	0.19	$1.0{\times}10^0$	2.15	$4.0{\times}10^{-1}$	1.19	$9.3{\times}10^{-1}$	–
`hsp90_kung`	13	3	0.19	$1.0{\times}10^0$	0.18	$1.0{\times}10^0$	0.83	$1.0{\times}10^0$	–
`cmet`	39	16	0.17	$1.0{\times}10^0$	0.36	$1.0{\times}10^0$	0.42	$1.0{\times}10^0$	–
`hsp90_single_ring`	8	2	0.16	$1.0{\times}10^0$	0.02	$1.0{\times}10^0$	0.24	$1.0{\times}10^0$	–
`hne`	23	7	0.13	$1.0{\times}10^0$	0.28	$1.0{\times}10^0$	0.07	$1.0{\times}10^0$	–
`urokinase`	4	1	0.13	$1.0{\times}10^0$	0.01	$1.0{\times}10^0$	0.25	$1.0{\times}10^0$	–
`jnk1`	29	7	0.12	$1.0{\times}10^0$	0.12	$1.0{\times}10^0$	1.19	$9.3{\times}10^{-1}$	–
`syk`	67	22	0.11	$1.0{\times}10^0$	0.20	$1.0{\times}10^0$	0.25	$1.0{\times}10^0$	–
`egfr`	7	3	0.07	$1.0{\times}10^0$	0.01	$1.0{\times}10^0$	0.02	$1.0{\times}10^0$	–
`irak4_s3`	4	1	0.07	$1.0{\times}10^0$	0.07	$1.0{\times}10^0$	0.05	$1.0{\times}10^0$	–
`jak2_set2`	12	5	0.06	$1.0{\times}10^0$	1.02	$1.0{\times}10^0$	0.49	$1.0{\times}10^0$	–
`bace1`	3	1	0.05	$1.0{\times}10^0$	1.09	$8.9{\times}10^{-1}$	0.43	$1.0{\times}10^0$	–
`dlk`	6	2	0.02	$1.0{\times}10^0$	0.06	$1.0{\times}10^0$	0.05	$1.0{\times}10^0$	–
`jak1`	7	2	0.02	$1.0{\times}10^0$	0.32	$1.0{\times}10^0$	0.10	$1.0{\times}10^0$	–
`bace_ciordia_prospective`	11	3	0.01	$1.0{\times}10^0$	0.07	$1.0{\times}10^0$	0.04	$1.0{\times}10^0$	–
`hsp90_woodhead`	4	1	0.01	$1.0{\times}10^0$	0.07	$1.0{\times}10^0$	0.15	$1.0{\times}10^0$	–

Two-sided self-calibration of the per-edge error bar (Figure S14). Seven arms are compared: the nominal test, which uses the reported standard errors unchanged; the two-sided correction at the pre-specified per-edge granularity, which corrects each edge by its own ratio; a coarser variant correcting each system by its aggregate ratio; a variant clipping the ratio to $[1/3, 3]$; a variant dividing the replicate standard deviation by the constant $c_4(3) = 0.886$, the expectation of the sample standard deviation at three draws, which widens the bars; a variant measuring the ratio on the two replicates the test does not read; and, at the same per-edge granularity, the one-sided correction that inflates only where the measurement says the bar is too tight, which is the direction of Table S9 applied per edge rather than per system. The constant $c_4(3)$ is unrelated to the per-edge calibration ratio c_e. Table S8 tabulates the four granularity-by-direction combinations.

Simulation protocols. The two protocols the main text names are specified here.

The pure-null simulation behind the realized levels. Every realized level quoted for a self-calibrated arm, in Section 6, in Section 8 and in the last column of Supplementary Table S8, is measured on a synthetic benchmark that carries no systematic error, so that every flag it produces is a false positive by construction. The topologies are the real ones: the same 48 systems the closure test admits, each with its own ligand-pair edge list and its own admission by the same rule (≥ 3 edges, ≥ 1 independent cycle), which depends on topology alone. The variances are the real ones: each edge keeps the per-edge reported standard error it carries, and that value is then declared to be the edge's true sampling standard deviation, which makes the reported bars perfectly calibrated by construction. The draw law is the null: every node potential is exact, so each edge's true $\Delta\Delta G$ is exactly the difference of its endpoints' potentials, and an observation is that value plus an independent Gaussian of the edge's own standard deviation. There is no systematic component anywhere and no correlation between edges. The replicate structure mirrors the benchmark's: three independent draws per edge, of which the first plays the role of replicate 0 and is the one the closure statistic is computed on, while all three are read by the arm's own calibration formula exactly as that arm reads the real replicates (the held-out arm reads only the second and third). Each arm's correction, tail probability and Benjamini–Hochberg step-up at $\alpha = 0.05$ are then the identical code paths used on the real data. The simulation is run for 400 draws, so a realized probability is measured to a Monte-Carlo standard error of at most $\sqrt{0.25/400} = 0.025$, and the generator is seeded once for the whole sweep, which makes the reported levels reproducible.

The reproducible-systematic magnitude used as ground truth. Detection comparisons need a label that does not come from the detector being scored, and the only one these data supply is reproduction across the three protocol repeats. It is defined as follows. Each replicate's network is fitted separately by the same generalized least squares, giving for replicate k and edge e a standardized residual $z_e^{(k)} = r_e^{(k)}/\sqrt{V_e^{\text{rep}}}$. The three signed residuals of an edge are averaged, $\bar{z}_e = \frac{1}{3}\sum_k z_e^{(k)}$, so that a residual reproducing in sign and magnitude survives the average while sampling noise cancels toward zero, and the edge's score is $|\bar{z}_e|$. A system's score is the median of $|\bar{z}_e|$ over its edges. This is the quantity the detectors of the head-to-head comparison are ranked against, by Mann–Whitney AUC of the binary flag against the per-system score. It uses only replicate residuals and no detector output, but it is not independent of the rules being scored, and the dependence is the reason no ordering is read out of that comparison (Section 6): $z_e^{(k)}$ divides by V_e^{rep}, the same per-edge reported variance the calibrated rule uses as its weight, so the label rewards a rule for standing close to that variance. "Edges whose sign and magnitude reproduce across independent replicates", wherever that phrase appears above, means edges with a large $|\bar{z}_e|$ in this sense.

Model architectures. The two learned models the main text compares against or uses as an amortized predictor are specified here in full.

Amortized predictor. The predictor used in the acquisition, stopping and reward experiments, referred to elsewhere in this article as the trunk, is small, and it is not an equivariant network. A ligand is featurized as a 1024-bit ECFP4 Morgan fingerprint (radius 2, chirality flag off) concatenated with six achiral RDKit descriptors: molecular weight, Crippen $\log P$, topological polar surface area, and counts of hydrogen-bond donors, hydrogen-bond acceptors and rotatable bonds. An edge is represented by the difference of its two endpoint feature vectors. A single parity-odd coordinate is appended, the difference of a summed signed volume over the assigned stereocentres of an ETKDGv3 embedding, giving 1031 features per edge; every other coordinate is invariant under reflection, so a readout without it would be constant on an enantiomer pair. The regressor is a bootstrap deep ensemble of eight multilayer perceptrons, each with two hidden layers of 64 units and SiLU activations and a scalar output, trained for 300 epochs on a squared-error objective with Adam at learning rate 5×10^{-3} and weight decay 10^{-4}, on inputs standardized by the training mean and standard deviation. Each member is seeded independently and fitted to its own bootstrap resample; the epistemic term is the spread across members, and the total σ combines it in quadrature with the aleatoric floor before a conformal rescaling.

Learned-variance foil. The learned head the calibrated bar is compared against is a separate and much smaller model from the amortized predictor above, and it is trained on the controlled panel rather than on the benchmark. Each training example is one simulated edge, from which six features are formed: the means and standard deviations of the forward and reverse work samples, the absolute difference of the two means, and the normalized overlap scalar. Including the overlap scalar gives the head the same overlap information the closed form uses, so the comparison is fair. The network is a multilayer perceptron with two hidden layers of 32 units and SiLU activations and two outputs, a mean and a log-variance, trained for 1500 steps of Adam at learning rate 5×10^{-3} on a Gaussian negative log-likelihood, with inputs standardized by the training mean and standard deviation. The realistic-budget arm trains on 200 edges of 20 work samples each; the large-budget arm raises that to 4000 edges, twenty times larger, and the corrected arms replace the objective by the β-NLL of Seitzer et al. [17] at $\beta = 0.5$ or average five independently seeded heads.

Granularity and direction of the self-calibrated null (Table S8). The self-calibrated null has two independent design choices, and the flag count depends on both. The *granularity* is whether an edge is corrected by its own replicate-measured ratio or by the ratio pooled over its whole system; per edge was pre-specified. The *direction* is whether the correction is applied only where the measurement says the bar is too tight, which can only remove flags, or in whichever direction the measurement points, which can also add them; the one-sided direction is the one the circularity concern of Section 6 calls for, and is the direction of Table S9. Table S8 reports all four combinations on the identical population, together with each arm's realized probability of at least one false flag under a pure-null simulation in which every flag is a false positive by construction. The count of surviving flags ranges from two to sixteen across the four, so no single self-calibrated flag list is a stable object, and none of the four arms is reported as a flag set. Only the pre-specified per-edge one-sided arm holds its level: the two-sided arms fire far above nominal because rescaling an edge

Table S8: **Self-calibrated cycle-closure flags by correction granularity and direction.** All four arms use the identical population (replicate 0, the 48 systems with at least one independent cycle) and the identical Benjamini–Hochberg level $\alpha = 0.05$; they differ only in whether the replicate-measured calibration ratio is applied per edge or pooled per system, and in whether it is applied only where it loosens a bar (one-sided) or in whichever direction the measurement points (two-sided). "published six" counts how many of the flags of Figure 7 the arm retains; "entering" lists the systems it adds. The last column is the realized probability of at least one false flag over 400 draws of a perfectly-calibrated synthetic null, against the nominal 0.05, with its exact binomial interval.

granularity	direction	flagged	published six	entering	null $P(\geq 1)$
reported se, uncorrected		6	all six	none	0.040 $[0.023, 0.064]$
per edge	one-sided	2	`bace`, `cdk8`	none	0.003 $[0.000, 0.014]$
per edge	two-sided	7	4 of 6	`irak4_s2`, `thrombin`, `tyk2`	0.588 $[0.538, 0.636]$
per system	one-sided	6	all six	none	not measured
per system	two-sided	16	all six	`ciordia_retro`, `cmet`, `galectin`, `irak4_s2`, `keranen_p2`, `mcl1`, `renin`, `thrombin`, `tnks2`, `tyk2`	0.240 $[0.199, 0.285]$

The per-edge two-sided arm retains `bace`, `brd4`, `cdk8` and `p38` and drops `faah` and `hif2a`. The per-edge one-sided arm loses more, retaining `bace` and `cdk8` alone; the two per-system arms retain all six. The per-system one-sided arm is the pre-registered circularity check of Table S9, whose harness reports the flag set rather than a synthetic-null level, so no realized probability is available for it; a one-sided correction can only widen bars, and the flag sets of both one-sided arms here are contained in the uncorrected one.

by its own replicate spread multiplies the statistic by σ^2/s^2, whose expectation $\nu_{\text{rep}}/(\nu_{\text{rep}} - 2)$ is undefined at the $\nu_{\text{rep}} = 2$ replicate degrees of freedom three replicates provide. Taken together, the two-sided arms bound how far a flag list can move under calibration error of the size actually present, and the one-sided per-edge arm gives the conservative count: two of the six survive being retested against their own measured width. The realized levels are measured over 400 draws of the pure-null simulation specified above, so a realized 0.003 is a single flag and is not resolved from zero.

QC sweep under the real learned-σ profile (Figure S15). Figure 7B's uniform $\times 0.15$ shrink is a stress test; here the learned head's measured, overlap-dependent miscalibration profile (Figure S2: ratio 0.09–0.20$\times$ across overlap 0.26–0.78) is propagated through the identical GLS + Benjamini–Hochberg pipeline, per edge, by percentile-rank transfer onto each of 1143 (of the 1145; the sole cycle-free system contributes the other two) real edges' own OpenFE `smallest_overlap` (an approximation: the two overlap statistics are on different scales, so only the profile's ordering and spread are transferred, not its raw values). The rank transfer approximates how the head's measured overconfidence would vary across these edges; it is not the head's own output on this benchmark. Four arms are run on the same 48-system, 1143-edge network. The calibrated sandwich ($\times 1$) flags 6/48 (12%); the real profile (primary) flags 42/48 (88%), 7.00$\times$ the calibrated count, past the pre-registered 2$\times$ threshold; a shuffled-profile control (the identical per-edge ratios with their association to overlap destroyed, seed 20260808) also flags 42/48; and the uniform $\times 0.15$ stress test again flags 42/48. The three shrinking arms are not identical sets of systems (real vs. shuffled and real vs. uniform each swap exactly one system, `bace1` for `egfr`), so the per-edge ratio assignment is heterogeneous. The shuffled-profile control, the identical multiset of per-edge ratios with the association to overlap destroyed, also flags 42 of 48: the profile's entire measured range (0.09–0.20$\times$) is severely overconfident, every ratio is at most 0.20, so even the mildest value inflates an edge's χ^2 contribution by at least $\sim$25×, and which systems cross the Benjamini–Hochberg threshold is set by the magnitude of the shrink rather than by its structure. The whole assignment is deterministic: each edge's transferred ratio is a fixed function of its own overlap percentile, the shuffled control draws its permutation from the recorded seed, and the fit at each arm is a closed-form recomputation, so re-running reproduces the four arm counts exactly.

Dose-response of the QC to σ miscalibration (Figure S16). Figure 7B and Supplementary Figure S15 each characterize the closure test at a small number of scale points (×1 calibrated, ×0.15 stress test / real profile); here a pre-registered, frozen grid of 16 uniform se scale factors (2.0–0.10×) is swept through the identical GLS + Benjamini–Hochberg pipeline on the same 48-system OpenFE network. Two readouts were frozen in advance: s_{onset}, the largest scale that flags strictly more systems than calibrated, and s_{50}, the largest scale at which at least half the systems are flagged. $s_{\text{onset}} = 0.9\times$ (7/48 vs. the calibrated 6/48) is exactly one system crossing the Benjamini–Hochberg threshold. That readout is grid-determined rather than a measurement of this benchmark: because the flagged count is monotone in the scale, a strictly-exceeds-calibrated onset returns whichever grid point below 1.0 sits just past the first system's threshold crossing. Independent probing places that system's crossing as high as $s = 0.995\times$, so a finer grid would return $s_{\text{onset}} = 0.995\times$, and in the continuum limit $s_{\text{onset}} \to 1.0\times$ for any benchmark with a system near the cutoff. It is reported as the pre-registered readout. The informative readout is magnitude-based, such as the largest scale that doubles the calibrated count, which here is 0.7× at 12/48. The substantive change is mid-curve: from 0.7× to 0.3× the flagged fraction climbs from 12/48 (25%) to 30/48 (62%), reaching half the benchmark at $s_{50} = 0.3\times$ (the true 50% crossing is bracketed by the adjacent 0.35× and 0.30× grid points, which flag 23/48 and 30/48 respectively). The learned head's measured, rank-transferred miscalibration band (0.09–0.20×; Figure S2, Supplementary Figure S15) lies entirely below s_{50}, so the head is not near the transition but well inside the saturated regime. The over-wide half of the grid is not flat either: relative to the calibrated 6/48, a bar only 30% too wide (1.3×) already loses five of the six flagged systems (1/48), and any wider bar (1.5×, 2.0×) flags none, so the requirement is calibration, not conservatism. Two consistency checks confirm that the sweep reproduces the published test path: the ×1.0 row reads 6/48, matching Figure 7A's calibrated count, and the ×0.15 row reads 42/48, matching Figure 7B and Supplementary Figure S15's uniform/real-profile arms. The response to the shrink magnitude is graded rather than threshold-like. The closure test is therefore selective over roughly a factor of three of σ under-estimation below the calibrated scale, and over-wide σ costs power in the other direction, so the calibrated scale is a two-sided optimum rather than a plateau. The sweep is deterministic: the sixteen scale factors are a frozen grid and the test at each of them is a closed-form recomputation on fixed inputs, so re-running reproduces the entire curve exactly.

Per-system flag robustness to each system's own calibration error (Table S9). The Figure 7 flag set raises a mild circularity concern: the per-system calibration of the sandwich null, measured independently against across-replicate spread (Figure 5), is heterogeneous, and four of the six flagged systems are themselves locally overconfident (`brd4` 0.76×, `faah` 0.90×, `bace` 0.96×, `hif2a` 0.96×; only `cdk8` 1.41× and `p38` 1.43× are conservative), so a flag could partly reflect that system's own too-tight bars rather than genuine physical inconsistency. That concern is tested directly. For replicate 0, over the 48 systems with ≥ 1 independent cycle (the identical population as Figure 7), each system's per-edge se is inflated by 1/ratio where its own ratio is below 1 (systems with ratio ≥ 1 are already conservative and are left unchanged), and the Benjamini–Hochberg test is recomputed over the whole set under this self-calibrated null. The criterion was fixed in advance:

Table S9: **Per-system reduced χ^2 before and after inflating each system's own per-edge standard error by its own measured calibration error.** "ratio" is the per-system root-mean-square ratio of reported standard error to across-replicate standard deviation of binding $\Delta\Delta G$, at $n = 3$. χ^2_{nom} is the nominal reduced χ^2; χ^2_{self} inflates that system's own standard error by 1/ratio where ratio < 1 and leaves it unchanged otherwise. Rows are the six flagged systems.

System	ratio	χ^2_{nom}	χ^2_{self}	Still flagged
`brd4`	0.76	15.40	8.97	yes
`bace`	0.96	5.57	5.13	yes
`faah`	0.90	3.71	2.99	yes
`cdk8`	1.41	2.68	2.68	yes (unchanged; already conservative)
`hif2a`	0.96	2.53	2.36	yes
`p38`	1.43	2.40	2.40	yes (unchanged; already conservative)

the result stands if every nominally-flagged system remains flagged, and is partial or negative if only some or none do. The population, the inflation rule and $\alpha = 0.05$ were fixed before this run. All six flags remain under their own self-calibrated null (Table S9), each self-calibrated reduced χ^2 (2.36–8.97) remaining far above the level a sampling-consistent network of this size and degree-of-freedom profile shows (Section 6, main text). This check pools the ratio over a whole system; the same one-sided inflation applied at the per-edge granularity this work pre-specified retains two of the six, and Table S8 reports both granularities in both directions. The distribution-free cross-replicate reproduction check of Figure 3B is a function of raw residual correlation only, never of se, so it does not depend on this null and is not affected by this concern in either direction. The per-system ratio in Table S9 is the root-mean-square reported standard error over the root-mean-square across-independent-replicate SD, pooled per system, not a naive per-system average of per-edge ratios or a naive ratio of arithmetic means, which for three of the six systems below would sit on the opposite side of ratio = 1. The ratios are raw n=3 SD ratios. Their residual small-sample bias is upward and target-specific, of order the reciprocal of the edge count, so debiasing would move all four of the locally overconfident flagged systems (`brd4` 0.76×, `faah` 0.90×, `bace` and `hif2a` 0.96×) further below one rather than toward it; the raw ratios used here are generous to those systems rather than the reverse. The computation is deterministic, and the full 48-system table is released with the analysis code.

The downstream audit: what is under test (Table S11). The test of Section 6 runs on a network's own residuals. Whether the same free calibration also improves downstream machine learning is a separate question, and the six tasks collected in Table S11 are the audit of it. Three performance tasks, interval sharpness, commit-to-synthesis selection and reward re-ranking over a measured congeneric set, run with the commit-trust decision task on the public OpenFF protein–ligand benchmark [10] (FEP+ and Merck data [16, 20], experimental $\Delta\Delta G$ labels, both-endpoint scaffold-disjoint out-of-distribution splits); edge ranking in active learning and the stopping decision run in controlled simulations. The amortized σ under test is the predictive uncertainty of the amortized $\Delta\Delta G$ predictor, the trunk specified above, on unseen $\Delta\Delta G$, $\sigma_{\text{total}} = \gamma\sqrt{\sigma^2_{\text{ens}} + \sigma^2_{\text{ale}}}$: deep-ensemble epistemic spread and benchmark aleatoric floor, conformally scaled at miscoverage $\alpha = 0.1$. It is *not* the per-edge BAR sampling standard error of Figure 4 or its learned-variance baseline of Figure S2; a bare "σ" anywhere in this audit means σ_{total}.

Sharpness, ranking and selection (Figures S18 and S19A). On real held-out residuals of that trunk at matched $\sim$0.90 marginal coverage the decomposed interval is slightly wider than flat split-conformal (width ratio 0.92×), because epistemic spread is small against the large irreducible out-of-distribution error; its conditional calibration is comparable to a fair Mondrian-by-target conformal and the advantage is not significant (ours−split -0.044 $[-0.121, 0.005]$). The synthetic illustration above, whose generative structure the decomposition knows, is up to 2.2× sharper than split-conformal; that does not transfer. For edge ranking, sandwich against naive weights differ by 0.98× $[0.93, 1.03]$ on calls-to-top-k (above): in a connected graph the GLS contrast mean is unbiased for any positive weights, so the weight value is second-order for ranking. So too for commit selection at matched volume, where the σ-aware ranking does not beat a raw-μ ranking (-0.006 $[-0.016, 0.007]$). A calibrated bar informs whether to act and not which action ranks first.

Commit trust against recalibrated baselines (Figure S19B). Fixing the ensemble mean and varying only the uncertainty in a commit rule (commit iff the lower-confidence bound $\mu - \kappa\sigma \geq \tau$), an overconfident epistemic-only σ over-claims badly: its realized-versus-claimed correctness shortfall is $+0.182$ $[0.132, 0.225]$ larger than the decomposed σ's. That number quantifies miscalibration; it does not rank the ways of curing it. On the same scale, temperature scaling [9] of that overconfident σ leaves $+0.034$ $[-0.018, 0.081]$. Split-conformal recalibration of it leaves -0.029 $[-0.077, 0.007]$. Both straddle zero, in opposite directions. Trustworthy decisions need calibration, and all three routes supply it. Neither does this task separate them on cost, since all three fit a scalar on the same held-out split; the closed form's cost advantage is the one Section 5 measures, a per-edge variance computed rather than learned, and none of these six tasks tests it.

Reward re-ranking over a measured congeneric set (Figure S17). Driving a trajectory-balance GFlowNet [4, 14] over `eg5` (the most σ-varying evaluable target), the calibrated lower-confidence-bound (LCB) reward lowers the true hit-rate, 0.140 against a raw predicted-mean reward's 0.249 (cal−raw -0.109), because on this set σ tracks chemical novelty ($\mathrm{corr}(\sigma_{\mathrm{total}}, \mathrm{affinity}) = +0.42$) rather than low quality. An LCB reward improves the hit-rate only where σ flags low-quality candidates (Table S11 below).

The stopping decision as a function of the assumed error (Table S10). Figure S20 reports one point of a curve. The experiment measures how the stopping decision depends on the error the learner assumes, everything else held fixed, and the whole dependence is tabulated here. The 0.15× arm of Figure S20 is a counterfactual stand-in: it imposes on the assumed error the miscalibration measured for a trained per-edge head in Figure S2, and is not an estimator run here. The two arms of the figure keep their own Monte-Carlo stream, so adding the sweep points leaves them unchanged.

The 0.94× row is the 6% under-estimate a same-budget estimator pooling single-label edges by overlap reaches on the identical labels the heads were given (Figure S2). At that scale the stopping decision is indistinguishable from the calibrated one, 28.4 calls against 29.6 at the same 0.82 correctness. Correctness at the stop first moves at 0.60×, and the halved budget of Figure S20B needs 0.20×. The contrast is therefore scoped, exactly as the quality-control collapse of the main text is scoped, to per-edge heads missing by 5–11×, and is not evidence that a same-budget learned estimator would stop early. The one effect that survives a fair comparison is in the opposite direction: a 6% over-estimate spends 30.8 calls for the same correctness, the decision cost of a conservative bar of Figure S19B.

Table S10: **Simulated stopping under a swept assumed error.** Assumed posterior standard error as a multiple of the true one, against oracle calls at the stopping rule, top-k correctness at that stop, and the mean absolute gap between claimed confidence and actual correctness over the budget; 60 seeds. Starred rows are the two arms of Figure S20.

assumed se / true se	calls at stop	correct at stop	mean |claimed − actual|
1.06× (conservative)	30.8	0.83	0.027
1.00× (calibrated)*	29.6	0.82	0.022
0.94×	28.4	0.82	0.020
0.80×	25.4	0.82	0.029
0.60×	21.1	0.77	0.059
0.40×	17.9	0.68	0.093
0.20×	14.9	0.60	0.127
0.15× (stand-in)*	14.4	0.60	0.137

The decision cost of a conservative bar (Figure S19B). Even a conservative bar carries a decision cost: a systematic ∼1.41× se inflation is itself a (safe-direction) miscalibration that, via the LCB rule, biases toward late stopping and under-commit. It errs in the safe direction against sampling noise, and because the independent-replicate SD is itself a lower bound on full reproducibility (Section 5), the inflation may be closer to neutral against the true run-to-run spread once pose and system-preparation variance are included.

Downstream audit summary (Table S11). Table S11 collects, for four downstream performance tasks (interval sharpness, active-learning edge weighting, commit-to-synthesis ranking, and reward re-ranking on the `eg5` congeneric set) and two decision tasks (commit trust and active-learning stopping), the calibrated method's effect relative to the relevant baseline. The top block shows no measurable performance gain (Figures S18, S3, S19A, S17 above). The bottom block shows what the two decision tasks are measured against: an overconfident σ and a counterfactual stand-in are both far worse, which measures the cost of miscalibration, while against a fair baseline, the same overconfident σ after temperature or split-conformal recalibration and a same-budget pooled estimator at its measured 0.94×, the effect is absent (Figures S19B, S20 and Table S10 above).

What the audit establishes. None of the six tasks puts the calibrated σ ahead of a fair baseline. On commit trust the physics σ matches post-hoc recalibration rather than beating it, and the stopping separation is absent at the six per cent a same-budget pooled estimator produces. The audit establishes a precondition rather than an advantage: decisions of this kind need calibration, an overconfident σ corrupts them, and the calibration demonstrably helps on real data inside the network, as the null of the test of Section 6, rather than downstream of it. The graph weight contributes the same way, through structure rather than magnitude, making the gauge directions exactly unidentifiable (Theorem S1) while by Gauss–Markov its value does not improve ranking.

Provenance of the experimental affinities. The grounding against measured affinity uses one ChEMBL target per system, verified to be a human single-protein target before use, and one assay type per system, taken in the order IC_{50} then K_i and never mixed within a system. A flagged system enters the guided-versus-random comparison only if at least 60% of its network ligands map to a measured value. The counts here are grounded ligands, which for two systems exceed the ligand count of the sub-network the measurement runs on by one: an edge enters only when

Table S11: **Effect of the decomposed σ relative to a baseline, across six downstream tasks.** Effects are the calibrated method minus the relevant baseline with a 95% bootstrap confidence interval; "real" denotes OpenFF benchmark out-of-distribution edges, "sim" denotes a controlled simulation. The two decision tasks are each given against an uncalibrated foil and against a fair baseline: the same σ recalibrated, and the assumed error a same-budget pooled estimator reaches.

Downstream task	Data	Effect (calibrated − baseline) [95% CI]	Verdict
Interval sharpness (out-of-distribution)	real	width split/decomposed 0.92×; cond. -0.044 $[-0.121, 0.005]$	no gain
Edge weighting (active learning)	sim	calls-to-top-k 0.98× $[0.93, 1.03]$	no gain
Commit ranking (matched volume)	real	precision -0.006 $[-0.016, +0.007]$	tie
Reward re-ranking (evaluable set, `eg5`)	real	true hit-rate -0.109 $[-0.150, -0.072]$	hurts (directional, single target)[a]
Commit trust vs overconfident σ	real	shortfall $+0.182$ $[0.132, 0.225]$ better	cost of miscalibration
Commit trust vs temperature-scaled σ	real	shortfall $+0.034$ $[-0.018, +0.081]$	tie
Commit trust vs split-conformal σ	real	shortfall -0.029 $[-0.077, +0.007]$	tie
Stopping vs 0.15× stand-in	sim	82% vs 60% correct at stop	cost of miscalibration
Stopping vs pooled estimator (0.94×)	sim	0.82 vs 0.82 correct, 29.6 vs 28.4 calls	tie

[a] The reward re-ranking effect is measured on a single evaluable target (`eg5`) and is directional, and it is not yet established as a general property of calibrated reward re-ranking across chemotypes.

both its endpoints carry an affinity, so a ligand all of whose partners are ungrounded drops out. That is `1_1W7H` on `p38` (35 grounded, 34 in the sub-network) and `CAT-4j` on `bace` (24 and 23); `cdk8` and `hif2a` lose none. The `eg5` grounding of Section 6 predates that coverage-gated join and was assembled differently, per ligand from one named ChEMBL assay rather than by a target-wide structure search, so its row records the assay as well as the target. Table S12 lists what was used.

Table S12: **ChEMBL provenance of the experimental affinities.** Target identifier, protein, assay type, and the number of network ligands with a measured value, for every system grounded against measured affinity: the four flagged systems that clear the 60% coverage requirement, and `eg5`, the clean system used for the comparison of accuracy against experiment. Affinities are converted at $RT \ln 10 = 1.364$ kcal/mol and aggregated per ligand by the median pChEMBL value.

system	ChEMBL target	protein	assay	ligands
`cdk8`	CHEMBL5719	cyclin-dependent kinase 8	IC_{50}	24
`hif2a`	CHEMBL1744522	endothelial PAS domain protein 1	IC_{50}	41
`p38`	CHEMBL260	mitogen-activated protein kinase 14	IC_{50}	35
`bace`	CHEMBL4822	beta-secretase 1	IC_{50}	24
`eg5`	CHEMBL4581	kinesin-like protein KIF11	IC_{50}	28

The `eg5` affinities were taken from a single named ChEMBL ATPase IC_{50} assay, identifier CHEMBL1101849, queried per ligand; the other four rows come from a target-wide connectivity-based structure search under the 60% coverage rule, for which no single assay identifier applies.

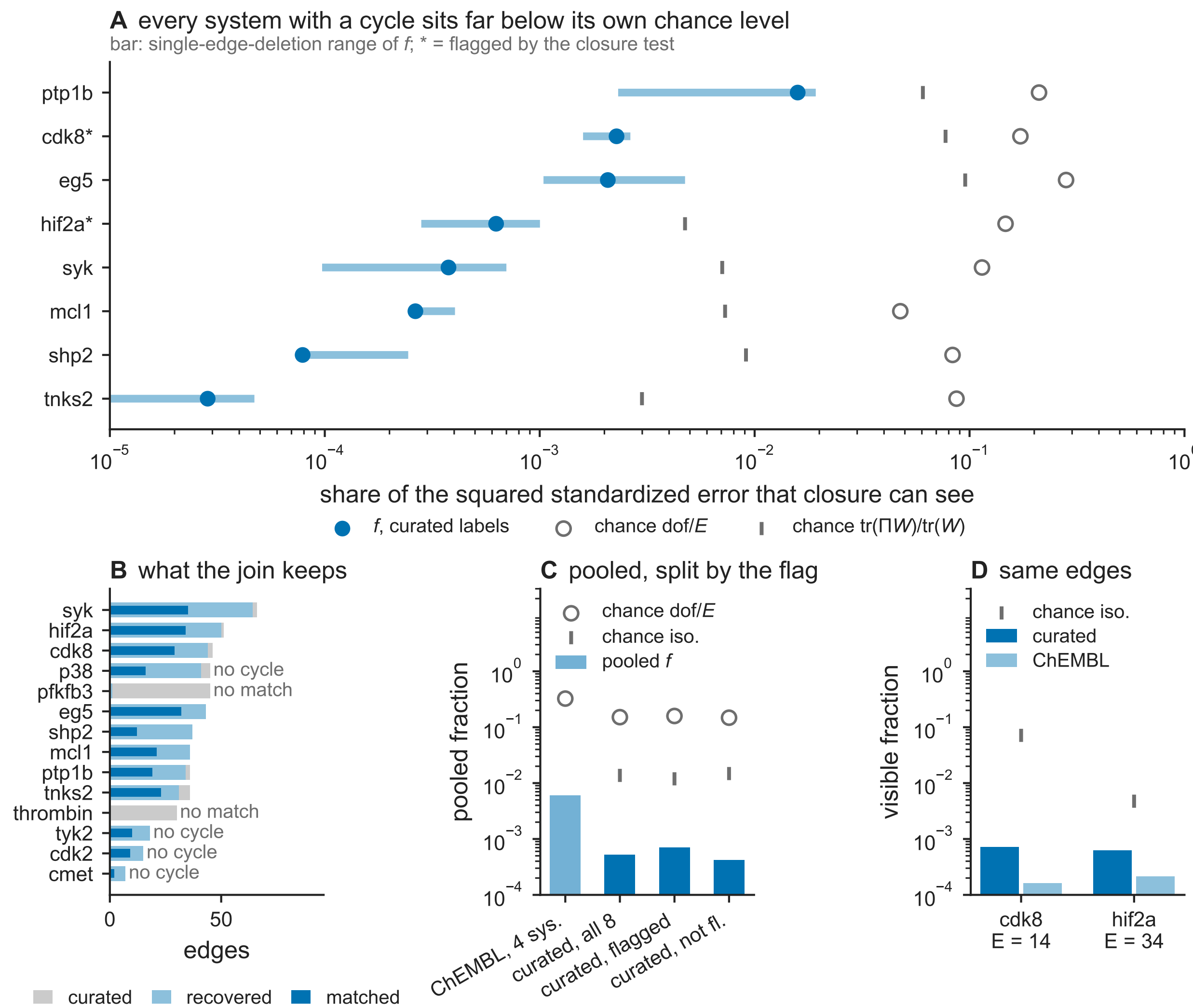

Figure S6: **The visible fraction of the error against curated experimental labels, over the benchmark.** (A) Per system with at least one independent cycle in its matched sub-network: the visible fraction f, the range it takes over single-edge deletions, and its two chance levels, dof$/E$ and the $\mathrm{tr}(\Pi W)/\mathrm{tr}(W)$ an error isotropic in kcal/mol would produce under the same projector. An asterisk marks a system the closure test flags. (B) What the join keeps, per system: curated edges offered, curated edges whose two ligands both recover into the network, and edges matched into the sub-network the measurement runs on, with the systems that carry no cycle or no matched edge named. (C) The pooled fraction and its chance levels for all eight measurable systems and for the flagged and unflagged systems separately, beside the four-system ChEMBL-grounded figure of Section 4, which is measured on different labels and a different edge set and is shown for scale rather than as a comparison. (D) The two label sources on identical edges, which holds the sub-network fixed: the noisier ChEMBL join gives the smaller fraction, as a gradient-field label error must.

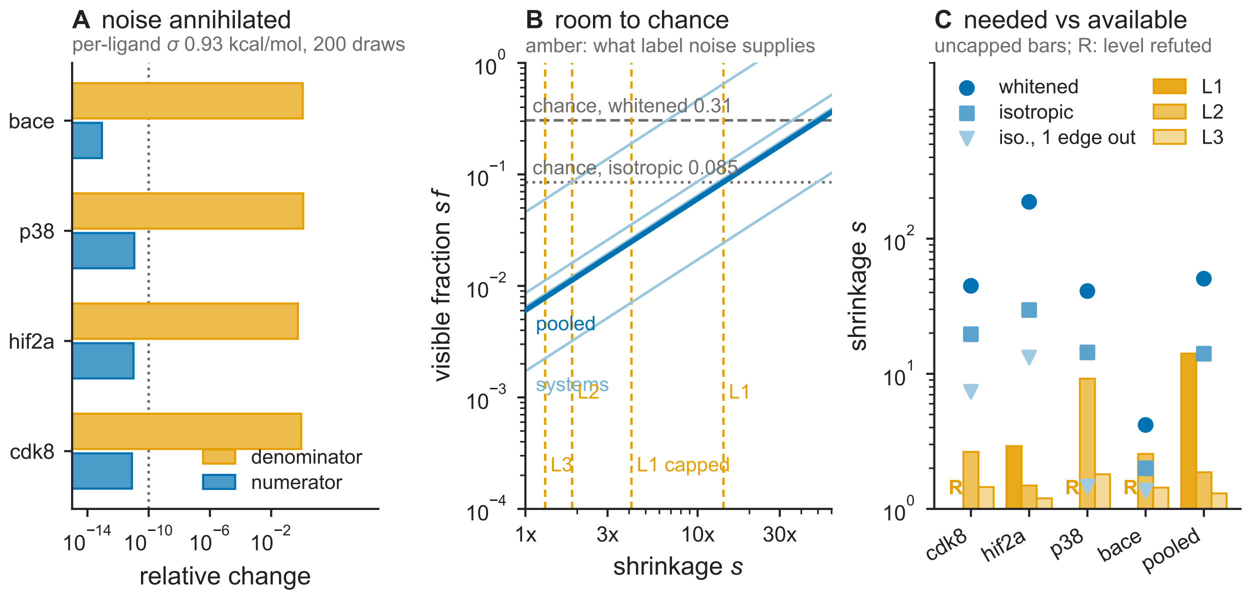

Figure S7: **The label-noise floor under the visible fraction.** (A) A synthetic per-ligand annotation error of standard deviation 0.93 kcal/mol is injected into each grounded network as the gradient field $\epsilon_e \to \epsilon_e + b_j - b_i$, 200 seeded draws per system: the worst relative change in the numerator $\|\Pi\tilde{\epsilon}\|^2$ over all draws, against the median relative growth of the denominator $\|\tilde{\epsilon}\|^2$, with the exactness tolerance fixed before the run marked. (B) The visible fraction under a denominator shrinkage s, per system and pooled, against the pooled whitened and isotropic chance levels; the vertical rules mark the shrinkage each declared label-noise level supplies. (C) Per system, the shrinkage needed to reach each chance level – whitened, isotropic, and isotropic under the least favourable single-edge deletion – against the shrinkage each level supplies, uncapped. An R marks a level that asserts more label variance than that system's whole measured error against experiment contains, and is therefore refuted by the data rather than truncated.

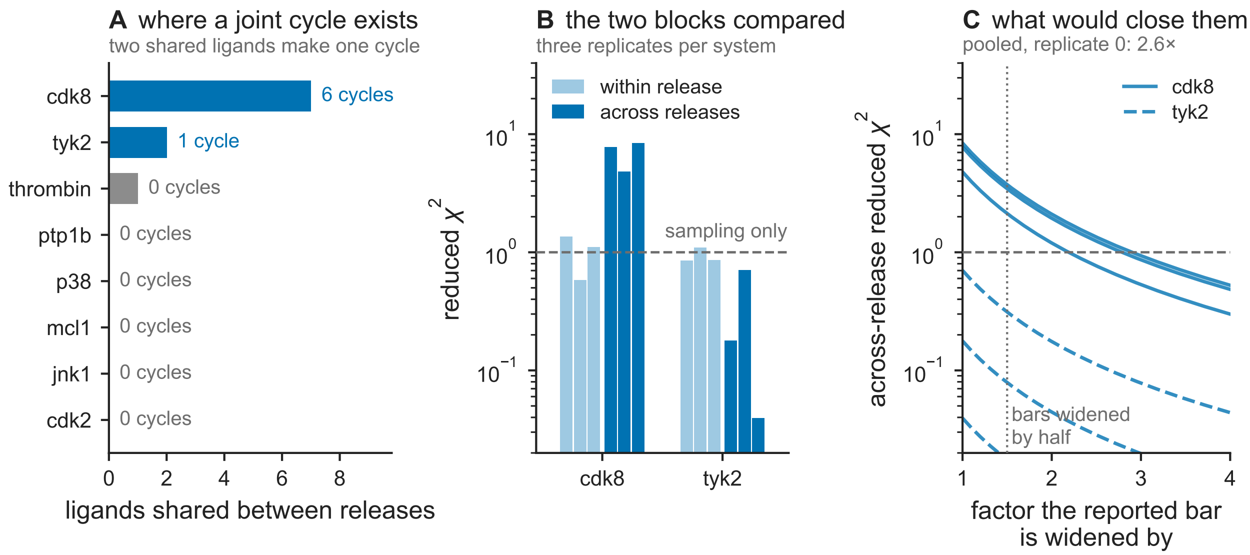

Figure S8: **What the multi-release systems can measure about between-preparation variance.** (A) For each system name that unions more than one separately released data set, the largest number of ligands any two of its releases share, annotated with the number of across-release cycles that creates. Two shared ligands are needed for one cycle. (B) Within-release and across-release reduced χ^2 for the two systems that carry an across-release cycle, three protocol repeats each, against the value sampling alone would give. (C) The across-release reduced χ^2 of the same two systems as every reported standard error is widened by a common factor, with the half-again widening of the main text's stress test marked; scaling every bar by λ divides the statistic by λ^2 exactly and leaves the fit unchanged. `cdk8`'s block measures a reuse of seven ligand names for different molecules between its two releases, not a disagreement between two preparations, and is not a between-preparation reading.

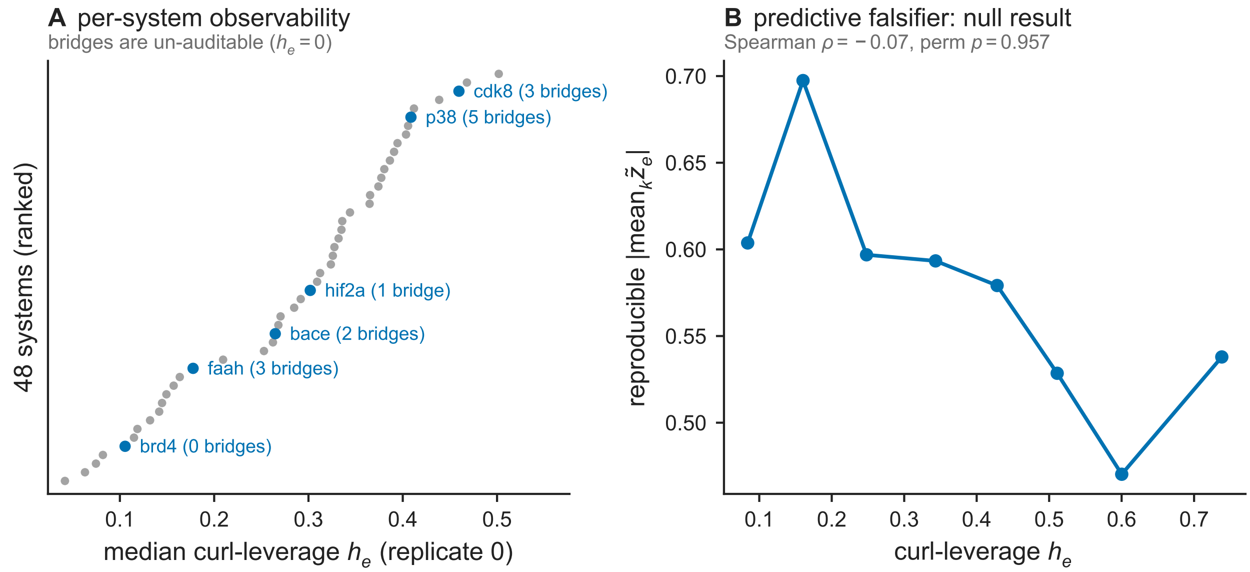

Figure S9: **Per-system curl-leverage, and curl-leverage versus reproducible residual magnitude.** (A) Median curl-leverage h_e per system, ranked, across the 48 systems with at least one independent cycle; the six cycle-closure-flagged systems are labeled with their number of bridge ($h_e = 0$) edges. (B) Curl-leverage h_e, binned into 8 quantiles, versus the mean studentized cross-replicate reproducible residual magnitude in each bin, with the Spearman correlation and permutation p-value.

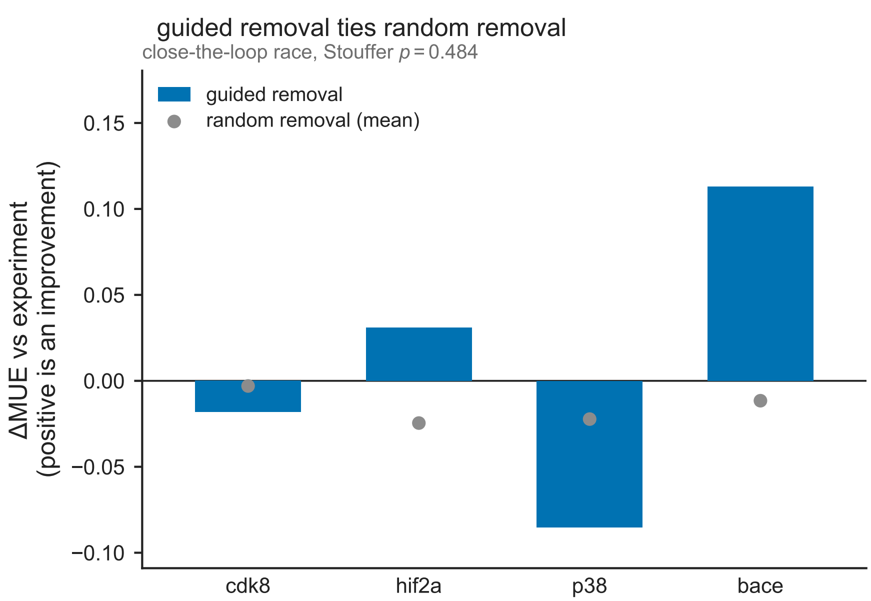

Figure S10: **Change in mean unsigned error against experimental affinity after guided versus random QC-edge removal.** Bars: change in mean unsigned error against experiment (ΔMUE) after guided removal of the QC-flagged high-$|z|$ edges, per system. Points: mean ΔMUE after a size-matched random removal, per system. Throughout, ΔMUE is MUE(full) – MUE(after), so a positive value is an improvement. Systems: `cdk8`, `hif2a`, `p38`, `bace`.

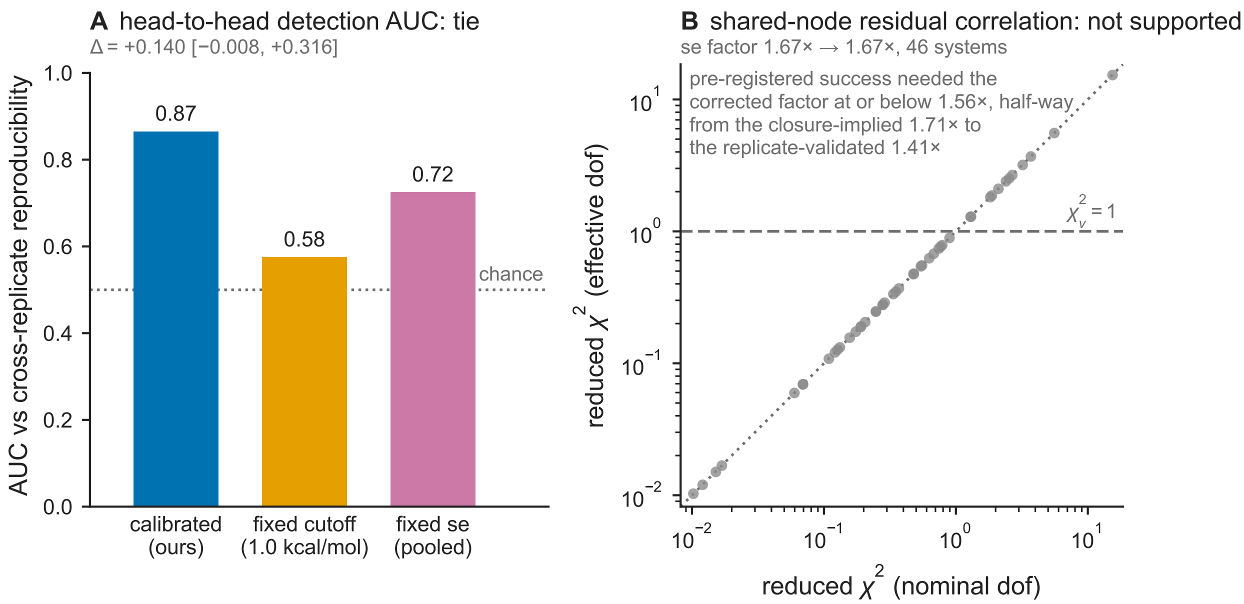

Figure S11: **Detection AUC of three flagging rules, and nominal versus correlation-corrected reduced χ^2.** (A) Area under the ROC curve (AUC) scored against the per-system reproducible-systematic magnitude defined in Computational details, for the calibrated per-edge null, a fixed 1.0 kcal/mol hysteresis cutoff, and a fixed-standard-error χ^2 test, with the chance level (AUC = 0.5) marked. (B) Per-system reduced χ^2 at the nominal degrees of freedom versus at the shared-node-correlation-corrected effective degrees of freedom, with the y=x line, the $\chi^2_\nu = 1$ reference, and the annotated inverted-median factor of 1.67×, the pre-registered functional evaluated over this panel's 46 systems.

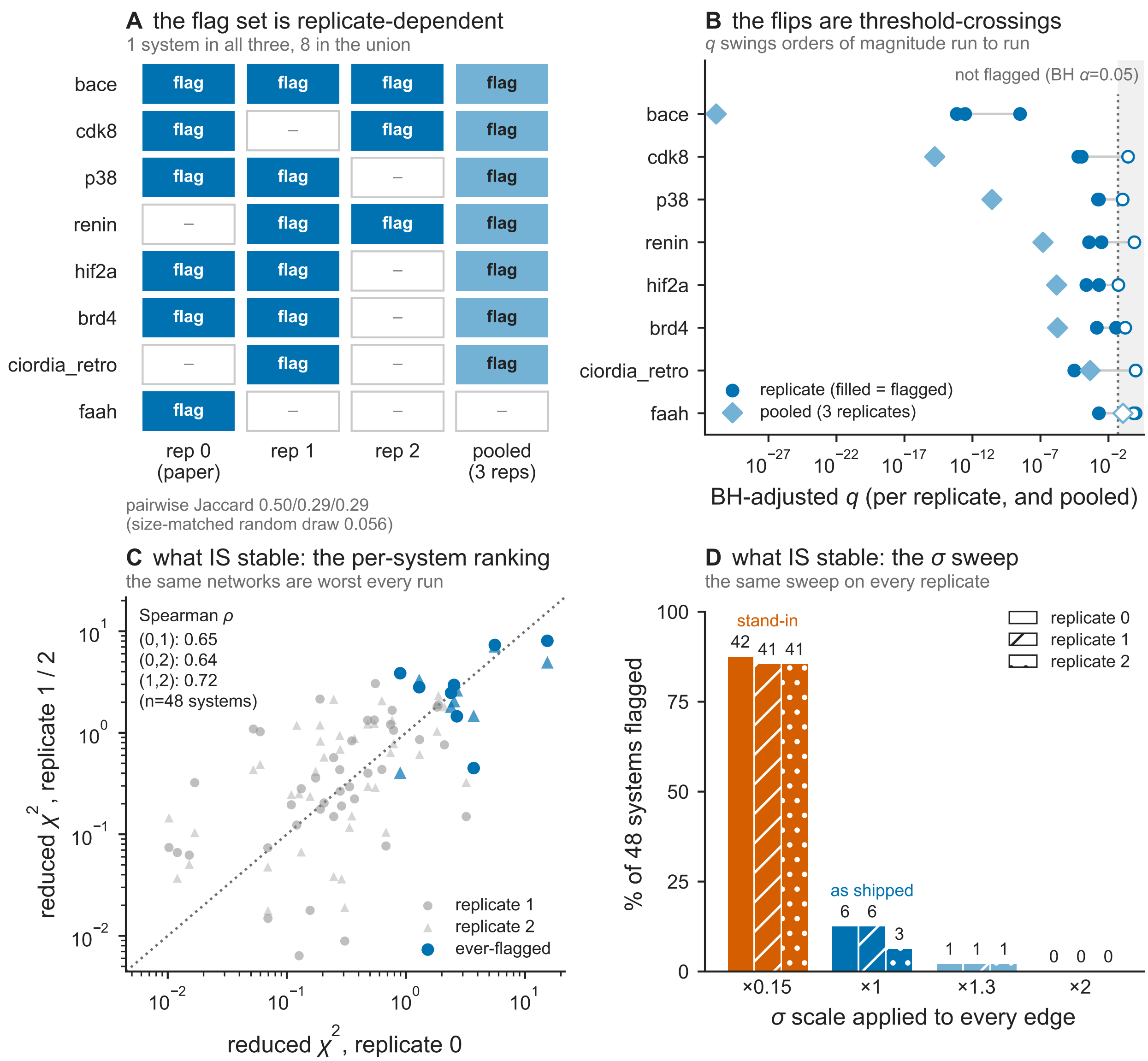

Figure S12: **Replicate stability of the flag set.** The Benjamini–Hochberg test at $\alpha = 0.05$ recomputed independently on each of the benchmark's three protocol repeats. Per-system adjusted p-values, which systems are flagged on which replicate, the pairwise Jaccard overlap of the three flag sets against a random-draw reference, the flag set obtained from the three replicates pooled, and the selectivity of the test across a swept uniform σ scale on each replicate.

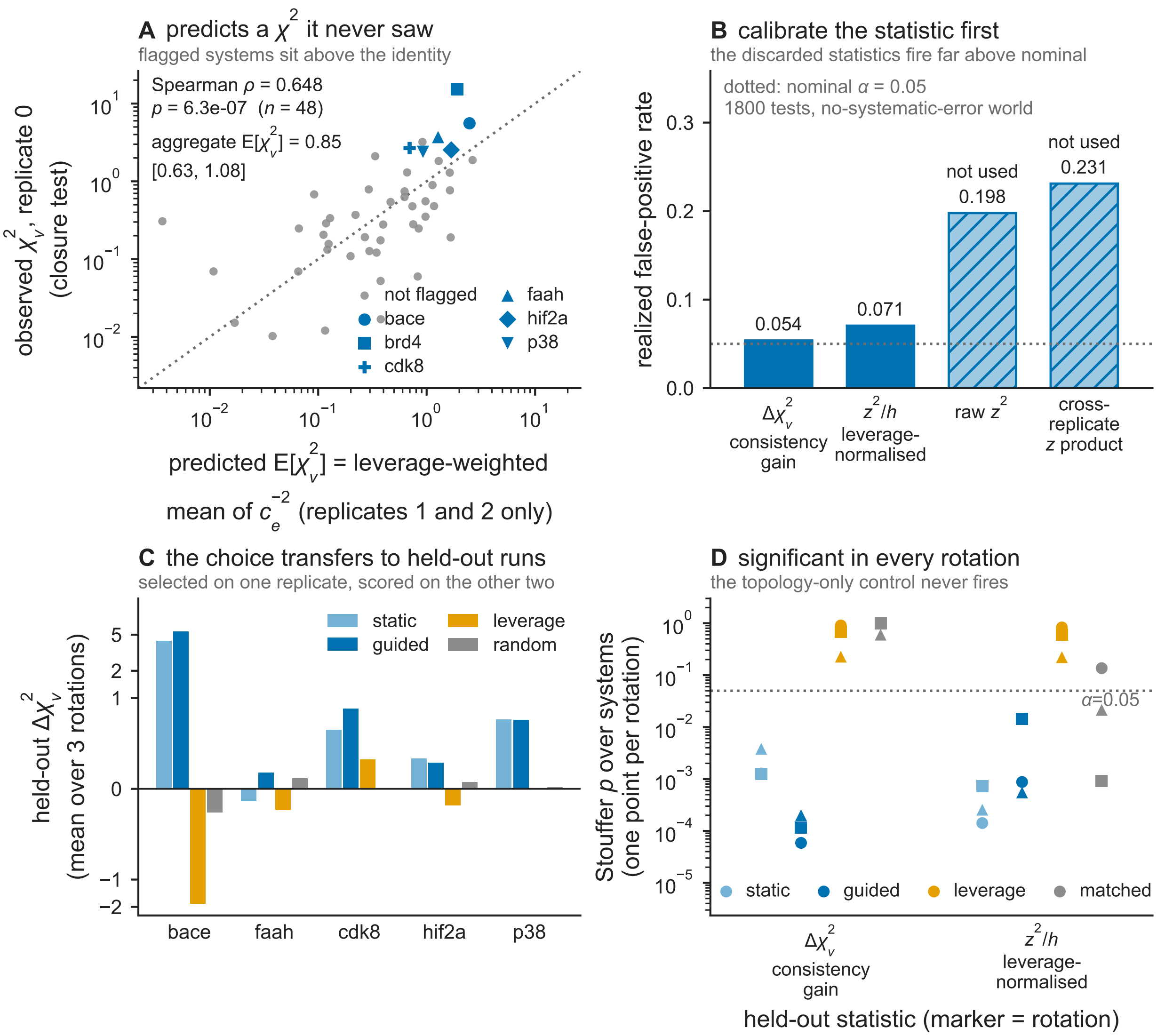

Figure S13: **Out-of-sample behaviour of the calibrated null.** (A) Per-system closure χ^2_ν predicted from the per-edge miscalibration measured on the two replicates not being predicted, against the observed value, with the identity line and the flagged systems marked. (B) Realized false-positive rate of each candidate held-out statistic under a simulation with no systematic error anywhere, against the nominal level, with the two statistics that fire far above nominal marked. (C) Held-out change in reduced χ^2 on the two replicates not used for selection, after removing the edges selected on the third, averaged over the three rotations and shown per testable flagged system, for a static and a guided selection rule, a topology-only rule built from curl-leverage, and equal-size random removal. (D) Stouffer-combined p-value over systems, one point per rotation, for each of the two statistics that held their nominal level, under the same selection rules together with an $|z|$-rank-matched control drawn from unflagged systems, with the $\alpha = 0.05$ line marked.

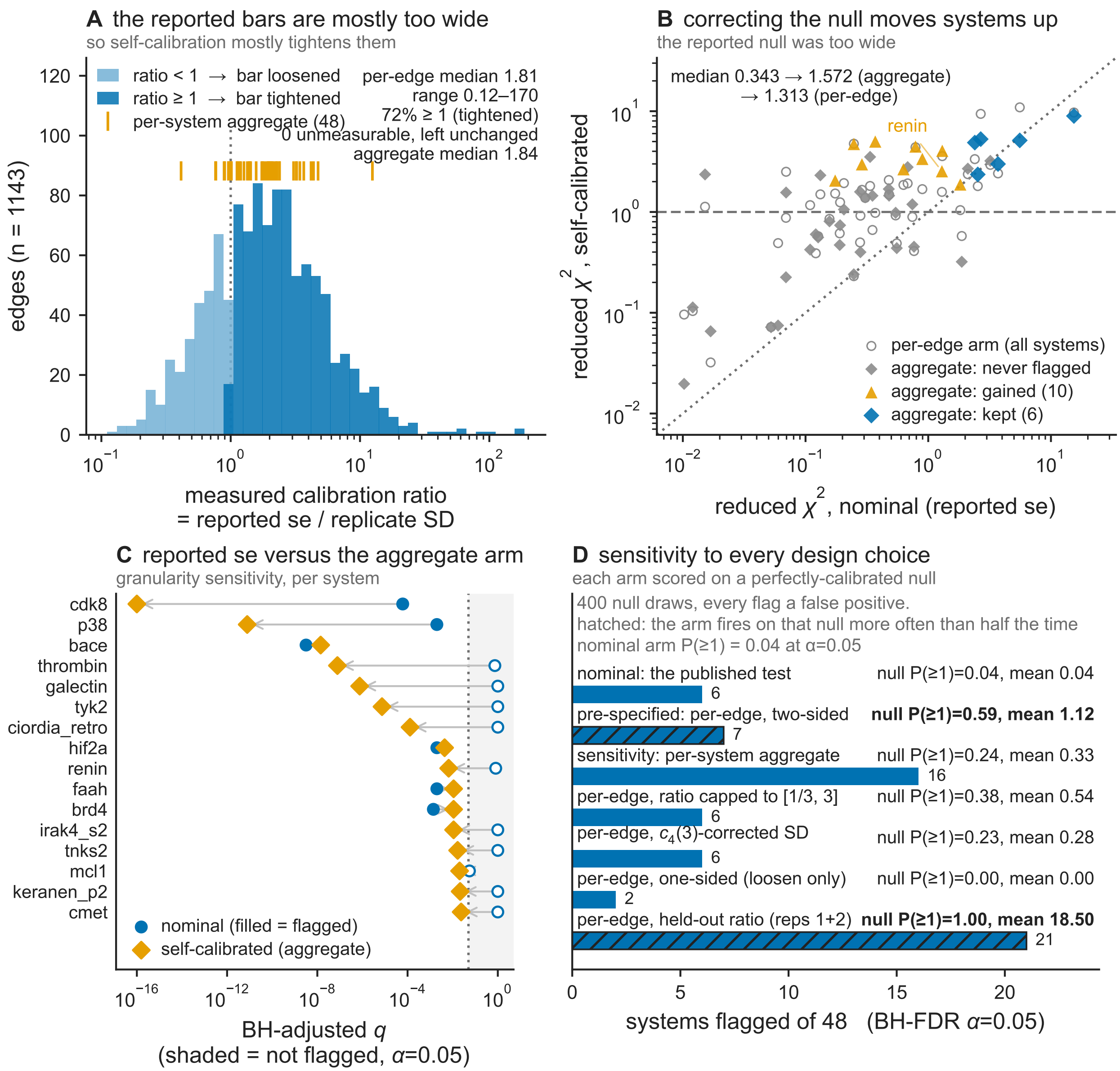

Figure S14: **Two-sided self-calibration of the per-edge error bar.** Per-system reduced χ^2 before and after replacing each system's reported standard error by its own replicate-measured calibration, which systems are gained and which are lost, and the realized probability of at least one false flag for each arm under a pure-null simulation, for the seven arms described in the text.

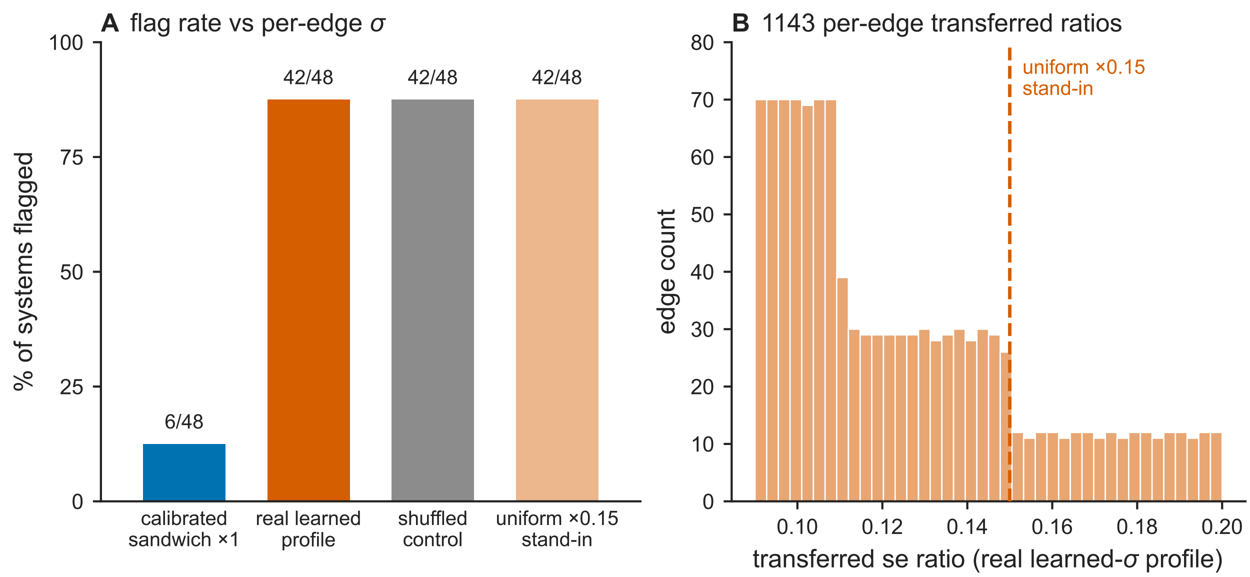

Figure S15: **Cycle-closure flagging under four per-edge σ assignments, and the distribution of the real-profile ratios.** (Left) Percentage of the 48-system OpenFE network flagged, for the calibrated sandwich, the learned head's measured overlap-dependent miscalibration profile transferred by percentile rank onto each edge's own overlap, a shuffled-profile control, and a uniform $\times 0.15$ shrink. (Right) Histogram of the 1143 per-edge transferred ratios from the real learned profile, with the uniform $\times 0.15$ stand-in marked.

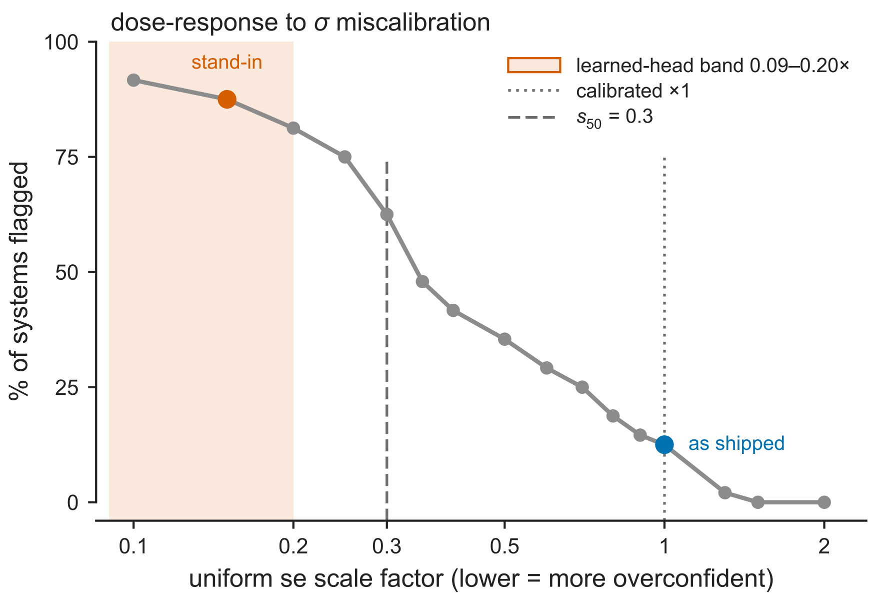

Figure S16: **Fraction of the OpenFE network flagged across a swept uniform σ scale factor.** Percentage of the 48-system network flagged by the cycle-closure test, across a frozen grid of 16 uniform standard-error scale factors from $2.0\times$ to $0.10\times$ (lower = more overconfident; $1.0\times$ is the calibrated sandwich). Vertical dotted line: the calibrated scale. Vertical dashed line: s_{50}, the largest scale at which at least half the systems are flagged. Shaded band: the learned head's measured, rank-transferred miscalibration range.

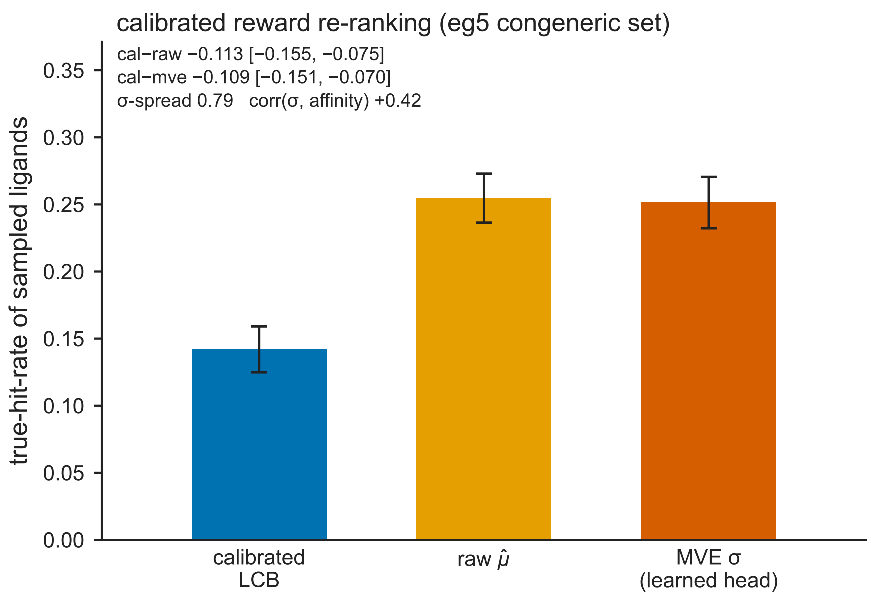

Figure S17: **True hit-rate of a trajectory-balance GFlowNet under three reward functions.** Mean true hit-rate of generated ligands, over 12 seeds with standard-error bars, for a calibrated lower-confidence-bound reward, a raw predicted-mean reward, and an overconfident MVE-uncertainty reward, on the `eg5` congeneric ligand series.

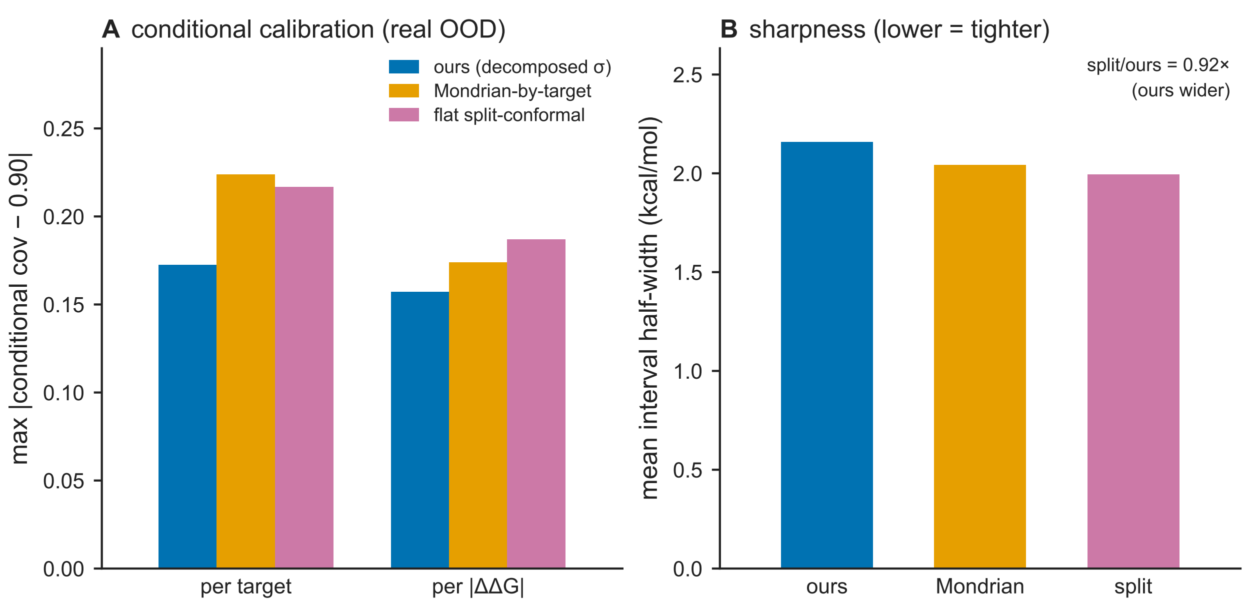

Figure S18: **Conditional calibration and interval sharpness on real out-of-distribution trunk residuals.** (A) Maximum conditional coverage error, computed per target and per $|\Delta\Delta G|$ bin, for the decomposed physics σ, a Mondrian-by-target conformal interval, and a flat split-conformal interval, each at approximately 90% marginal coverage. (B) Mean prediction-interval half-width for the same three methods.

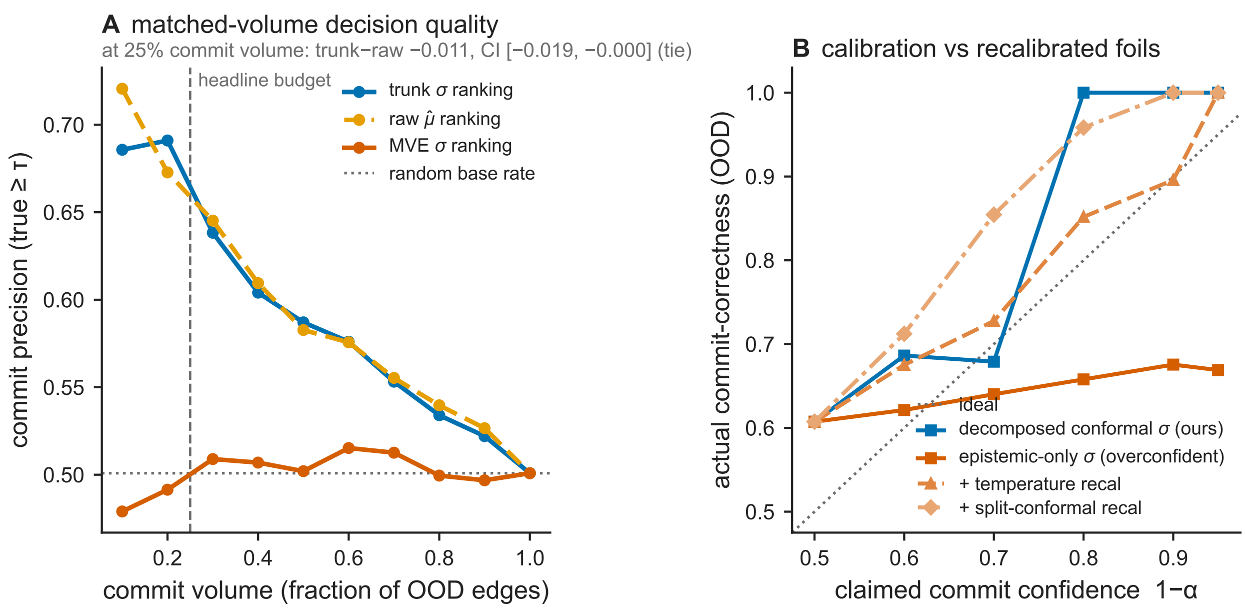

Figure S19: **Commit-to-synthesis ranking and calibration on real out-of-distribution edges.** (A) Commit precision (fraction of committed candidates with true value at or above a threshold τ) versus commit volume (fraction of edges committed), for candidates ranked by the trunk σ-aware margin, by the raw predicted mean, and by a learned MVE-head margin, with the random base rate marked. (B) Actual commit-correctness versus claimed commit confidence $1-\alpha$, for the decomposed physics σ, an epistemic-only σ, and the same epistemic-only σ after temperature and after split-conformal recalibration, with the ideal (actual = claimed) line.

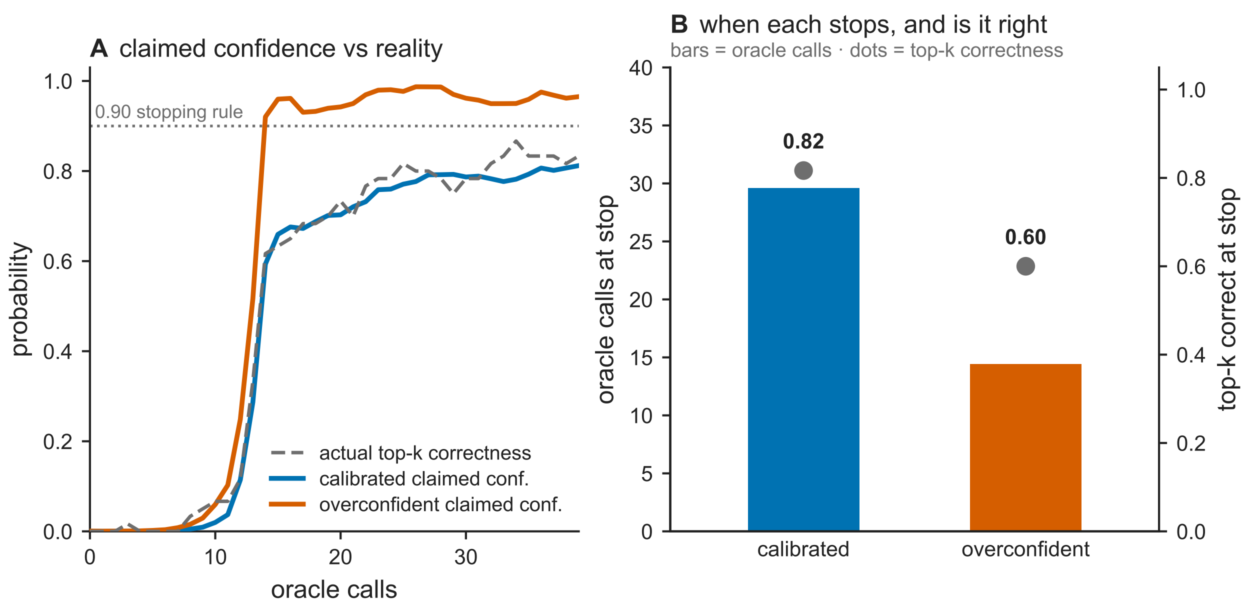

Figure S20: **Simulated active-learning stopping under calibrated versus counterfactually overconfident claimed uncertainty.** (A) Actual top-k correctness and claimed stopping confidence, averaged over 60 seeds, versus the number of oracle calls, for a calibrated learner and for a learner whose assumed posterior standard error is scaled by 0.15×. (B) Oracle calls at the stopping rule (bars) and top-k correctness at that stop (points), for the two learners.

References

[1] Matteo Aldeghi, Alexander Heifetz, Michael J. Bodkin, Stefan Knapp, and Philip C. Biggin. Accurate calculation of the absolute free energy of binding for drug molecules. *Chemical Science*, 7(1):207–218, 2016. doi: 10.1039/c5sc02678d.

[2] Maximilian Balandat, Brian Karrer, Daniel R. Jiang, Samuel Daulton, Benjamin Letham, Andrew Gordon Wilson, and Eytan Bakshy. BoTorch: A Framework for Efficient Monte-Carlo Bayesian Optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[3] Hannah M. Baumann, Joshua T. Horton, Michael M. Henry, Alyssa Travitz, Benjamin Ries, Richard J. Gowers, David W. H. Swenson, Iván Pulido, Dominic Rufa, David L. Dotson, Nupur Bansal, Joseph P. Bluck, Howard Broughton, Kira Campbell, Lili Cao, Benedikt Frieg, Vytautas Gapsys, Hendrik Göddeke, Marco Klähn, Sirish Kaushik Lakkaraju, Stephanie M. Linker, Thomas Löhr, Aniket Magarkar, Sergio Pérez-Conesa, Hans E. Purkey, Hayk Saribekyan, Jenke Scheen, Christina E. M. Schindler, Thomas Steinbrecher, Chaya D. Stern, Patricia Suriana, William C. Swope, Gary Tresadern, Lev Tsidilkovski, Binqing Wei, Alexander H. Williams, Yao Wu, Ivy Zhang, John D. Chodera, James R. B. Eastwood, David L. Mobley, and Irfan Alibay. Large-Scale Collaborative Assessment of Binding Free Energy Calculations for Drug Discovery Using OpenFE. *Journal of Chemical Information and Modeling*, 66(11):6429–6452, 2026. doi: 10.1021/acs.jcim.6c00089.

[4] Emmanuel Bengio, Moksh Jain, Maksym Korablyov, Doina Precup, and Yoshua Bengio. Flow Network based Generative Models for Non-Iterative Diverse Candidate Generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[5] Yoav Benjamini and Daniel Yekutieli. The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29(4), 2001. doi: 10.1214/aos/1013699998.

[6] Panagis Filippakopoulos, Jun Qi, Sarah Picaud, Yao Shen, William B. Smith, Oleg Fedorov, Elizabeth M. Morse, Tracey Keates, Tyler T. Hickman, Ildiko Felletar, Martin Philpott, Shonagh Munro, Michael R. McKeown, Yuchuan Wang, Amanda L. Christie, Nathan West, Michael J. Cameron, Brian Schwartz, Tom D. Heightman, Nicholas La Thangue, Christopher A. French, Olaf Wiest, Andrew L. Kung, Stefan Knapp, and James E. Bradner. Selective inhibition of BET bromodomains. *Nature*, 468(7327):1067–1073, 2010. doi: 10.1038/nature09504.

[7] Peter I. Frazier, Warren B. Powell, and Savas Dayanik. A Knowledge-Gradient Policy for Sequential Information Collection. *SIAM Journal on Control and Optimization*, 47:2410–2439, 2008. doi: 10.1137/070693424.

[8] Jacob R. Gardner, Geoff Pleiss, David Bindel, Kilian Q. Weinberger, and Andrew Gordon Wilson. GPyTorch: Blackbox Matrix-Matrix Gaussian Process Inference with GPU Acceleration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[9] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On Calibration of Modern Neural Networks. In *International Conference on Machine Learning (ICML)*, 2017.

[10] David Hahn, Christopher Bayly, Melissa L. Boby, Hannah Bruce Macdonald, John Chodera, Vytautas Gapsys, Antonia Mey, David Mobley, Laura Perez Benito, Christina Schindler, Gary Tresadern, and Gregory Warren. Best Practices for Constructing, Preparing, and Evaluating

Protein-Ligand Binding Affinity Benchmarks [Article v1.0]. *Living Journal of Computational Molecular Science*, 4:1497, 2022. doi: 10.33011/livecoms.4.1.1497.

[11] Tuomo Kalliokoski, Christian Kramer, Anna Vulpetti, and Peter Gedeck. Comparability of mixed IC_{50} data – a statistical analysis. *PLoS ONE*, 8(4):e61007, 2013. doi: 10.1371/journal.pone.0061007.

[12] D. J. Klein and M. Randić. Resistance distance. *Journal of Mathematical Chemistry*, 12:81–95, 1993. doi: 10.1007/BF01164627.

[13] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[14] Nikolay Malkin, Moksh Jain, Emmanuel Bengio, Chen Sun, and Yoshua Bengio. Trajectory balance: Improved credit assignment in GFlowNets. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[15] Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized Quantile Regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[16] Christina E. M. Schindler, Hannah Baumann, Andreas Blum, Dietrich Böse, Hans-Peter Buchstaller, Lars Burgdorf, Daniel Cappel, Eugene Chekler, Paul Czodrowski, Dieter Dorsch, Merveille K. I. Eguida, Bruce Follows, Thomas Fuchß, Ulrich Grädler, Jakub Gunera, Theresa Johnson, Catherine Jorand Lebrun, Srinivasa Karra, Markus Klein, Tim Knehans, Lisa Koetzner, Mireille Krier, Matthias Leiendecker, Birgitta Leuthner, Liwei Li, Igor Mochalkin, Djordje Musil, Constantin Neagu, Friedrich Rippmann, Kai Schiemann, Robert Schulz, Thomas Steinbrecher, Eva-Maria Tanzer, Andrea Unzue Lopez, Ariele Viacava Follis, Ansgar Wegener, and Daniel Kuhn. Large-Scale Assessment of Binding Free Energy Calculations in Active Drug Discovery Projects. *Journal of Chemical Information and Modeling*, 60:5457–5474, 2020. doi: 10.1021/acs.jcim.0c00900.

[17] Maximilian Seitzer, Arash Tavakoli, Dimitrije Antic, and Georg Martius. On the pitfalls of heteroscedastic uncertainty estimation with probabilistic neural networks. In *International Conference on Learning Representations (ICLR)*, 2022.

[18] Michael R. Shirts and John D. Chodera. Statistically optimal analysis of samples from multiple equilibrium states. *The Journal of Chemical Physics*, 129(12):124105, 2008. doi: 10.1063/1.2978177.

[19] Alexander D. Wade, Agastya P. Bhati, Shunzhou Wan, and Peter V. Coveney. Alchemical Free Energy Estimators and Molecular Dynamics Engines: Accuracy, Precision, and Reproducibility. *Journal of Chemical Theory and Computation*, 18(6):3972–3987, 2022. doi: 10.1021/acs.jctc.2c00114.

[20] Lingle Wang, Yujie Wu, Yuqing Deng, Byungchan Kim, Levi Pierce, Goran Krilov, Dmitry Lupyan, Shaughnessy Robinson, Markus K. Dahlgren, Jeremy Greenwood, Donna L. Romero, Craig Masse, Jennifer L. Knight, Thomas Steinbrecher, Thijs Beuming, Wolfgang Damm, Ed Harder, Woody Sherman, Mark Brewer, Ron Wester, Mark Murcko, Leah Frye, Ramy Farid, Teng Lin, David L. Mobley, William L. Jorgensen, Bruce J. Berne, Richard A. Friesner, and Robert Abel. Accurate and Reliable Prediction of Relative Ligand Binding Potency in

Prospective Drug Discovery by Way of a Modern Free-Energy Calculation Protocol and Force Field. *Journal of the American Chemical Society*, 137:2695–2703, 2015. doi: 10.1021/ja512751q.

[21] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2002. doi: 10.1145/775047.775151.

[22] Barbara Zdrazil, Eloy Felix, Fiona Hunter, Emma J. Manners, James Blackshaw, Sybilla Corbett, Marleen de Veij, Harris Ioannidis, David Mendez Lopez, Juan F. Mosquera, Maria Paula Magarinos, Nicolas Bosc, Ricardo Arcila, Tevfik Kiziloren, Anna Gaulton, A. Patricia Bento, Melissa F. Adasme, Peter Monecke, Gregory A. Landrum, and Andrew R. Leach. The ChEMBL Database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1):D1180–D1192, 2024. doi: 10.1093/nar/gkad1004.